

NATIONAL TRANSPORTATION SAFETY BOARD

Investigative Hearing

US Army PAT25, Sikorsky UH-60L and

PSA Airlines flight 5342

MHI RJ Aviation CL-600-2C10 (CRJ700), N709PS

Midair Collision in Washington, DC

January 29, 2025

Docket No.	SA-544
EXHIBIT	
2-HELO-A	

Helicopter Operations and Human Performance Factual Report

(99 Pages)

National Transportation Safety Board

Office of Aviation Safety

Washington, DC 20594



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HELICOPTER OPERATIONS AND HUMAN PERFORMANCE

Group Chairs' Factual Report

June 13, 2025

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ACCIDENT

Location: Washington, District of Columbia

Date: January 29, 2025

Time: 2048 EST ¹
0148 UTC

Helicopter: UH60-L / SNL 00-26860 / Sikorsky Build No. 702614

A HELICOPTER OPERATIONS AND HUMAN PERFORMANCE GROUP

Group Co-Chair William Bramble, Ph.D.
National Transportation Safety Board
Washington, DC

Group Co-Chair Van McKenny
National Transportation Safety Board
Washington, DC

Group Member Mark Ward
National Transportation Safety Board
Anchorage, AK

Group Member David Blair
Sikorsky Aircraft
Stratford, CT

Group Member Paul Kim
PSA Airlines
Washington, DC

Group Member CW5 Nicholas Jacoby
US Army, The Army Aviation Brigade
Ft Belvoir, VA

B SUMMARY

On January 29, 2025, about 2048 eastern standard time (EST), a Sikorsky UH-60L operated by the US Army under the callsign PAT25, tail number 00-26860, and PSA Airlines flight 5342, a MHI [Mitsubishi Heavy Industries] RJ Aviation (Bombardier) CL-600-2C10 (CRJ700), N709PS, collided in flight approximately 0.5 miles southeast

¹ All subsequent times are in eastern standard time unless otherwise indicated.
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of Ronald Reagan Washington National Airport (DCA), Arlington, Virginia, and impacted the Potomac River in southwest Washington, DC.

The 2 pilots, 2 flight attendants, and 60 passengers aboard the airplane and all 3 crew members aboard the helicopter were fatally injured. Both aircraft were destroyed by impact forces. Flight 5342 was a scheduled domestic passenger flight operating under the provisions of 14 Code of Federal Regulations Part 121 from Wichita Dwight D. Eisenhower National Airport (ICT), Wichita, Kansas, to DCA. The Army helicopter originated from Davison Army Airfield (DAA), Fort Belvoir, Virginia, and one of the pilots was conducting an annual check ride flight with night-vision goggles (NVGs) with the other pilot. Night visual meteorological conditions prevailed at the time of the accident.

C GROUP ACTIVITIES

The helicopter operations and human performance group formed on January 31, 2025. Between February 1 and February 13, 2025, the group obtained and reviewed various flight crew and operational records. The group conducted four interviews with next of kin and 15 interviews with US Army personnel. The group examined helicopter wreckage, and documented cockpit controls and displays. The group identified, inventoried and secured crew items and night vision goggle cases and components. Group members auditioned the helicopter cockpit voice recorder (CVR), facilitated the laser scanning of a UH-60L by NTSB vehicle performance specialists, and documented 12th Aviation Battalion facilities. The group completed its on-scene field notes February 14, 2025.

Between February 28 and May 5, 2025, the group conducted four additional next of kin interviews and four additional US Army personnel interviews. The group organized and conducted two simulator observation activities in a US Army Blackhawk Aircrew Trainer (BAT) at Ft. Belvoir. The group coordinated aerial observation flights involving a US Army Blackhawk and a PSA CL-600 (in support of a visibility study conducted by NTSB vehicle performance specialists). The group continued to collect NVG components as they were recovered by divers, and it performed a layout and examination of crew helmets and NVG components. The group obtained documents pertaining to US Army helicopter operations and US Army aviation safety programs and processes.

D DETAILS OF THE INVESTIGATION

1.0 History of flight

PAT 25 carried a crew of three, an instructor pilot (IP) who also served as pilot in command, a pilot (referred to as the "PI") who was receiving an NVG annual

proficiency and readiness test (APART), and a crew chief (CE). All three were assigned to B Company, 12th Aviation Battalion.

The IP printed a completed preflight risk assessment called a Risk-Common Operational Picture (R-COP) at 1216. The battalion flight schedule listed the crew's planned crew briefing time as 1400, with a scheduled takeoff at 1800.² The R-COP indicated the PI would be sitting in the left pilot seat and the IP would be sitting in the right pilot seat.³ It also indicated that the flight was scheduled to conclude at 2100 with an after-action review at 2130. The route listed on the R-COP was Davison Army Airfield (DAA) - Quantico Marine Corps Air Facility (NYG) - The Greenhouse Airport (02VA) - Manassas Regional Airport (HEF) - Laytonsville Federal Support Center Heliport (1MD5) - DC HELO - Maryland Airport (2W5) - DAA. The flight plan filed with DAA Base Operations (Attachment 4) was: Direct Culpeper Regional Airport (KCJR) - Leesburg Executive Airport (KJYO) - DC HELO - DAA Traffic Patterns. The flight was to be operated within the Washington DC Special Flight Rules Area (SFRA) and Flight Restricted Zone (FRZ), or just west of those areas. DC Helicopter Routes 1 and 4 would be used to transit south through the core of DCA Class B airspace. Figure 1 provides a depiction of the planned route, as filed. The depiction was created by investigators using ForeFlight software on an Apple iPad.

The mission packet (Attachment 4), a kneeboard sized preflight document that contained mission specific information for quick reference by aircrew, indicated that one of the pilots printed a helicopter performance planning card at 1305.⁴ It indicated that the preflight inspection of the helicopter was 1430. Interviews indicate that the IP approached a battalion standardization pilot who was also a mission briefing officer (MBO) in the B Company hangar that afternoon for a required briefing.⁵ The IP told the MBO he intended to perform a time-on-target mission and land at a private airfield.⁶ After that, he intended to fly to the Frederick, Maryland area, land at another LZ, and then proceed south through the Washington, DC helicopter routes to DAA where he and the PI would perform emergency procedures. The R-COP risk rating he had calculated for the training mission was low.⁷ The MBO briefed the PI by instructing him to review the landing zone and pickup zone (LZ and PZ) cards for his intended stops. The MBO added handwritten notes to the R-COP that stated, "Review site file with crew for LZ", "Touchdown EPs at DAA", and "Discuss

² Risk Common Operating Procedure (R-COP), a detailed risk assessment worksheet created by the pilot in command, reviewed and approved by authorized personnel.

³ A battalion standardization pilot told investigators it was very common for a CE to sit at the CE crew station behind the person being evaluated on such flights. (Interview with CW4 Matthew Ingalls, Battalion Standardization Pilot p. 31)

⁴ Attachment X: Photos of recovered mission packet.

⁵ Interview with CW4 Matthew Ingalls, Battalion Standardization Pilot p. 7.

⁶ An LZ card for this field was included in the mission packet.

⁷ See section 6.2 for detailed description of the R-COP flight risk assessment tool.

ERM⁸ and 1070 as... (illegible)"⁹. The MBO signed the R-COP (Attachment 4).¹⁰ The MBO told investigators the IP and the PI likely intended to wear NVGs throughout the entire flight, because if they planned to remove them for some portion of the flight, the IP was required to inform him about it.¹¹

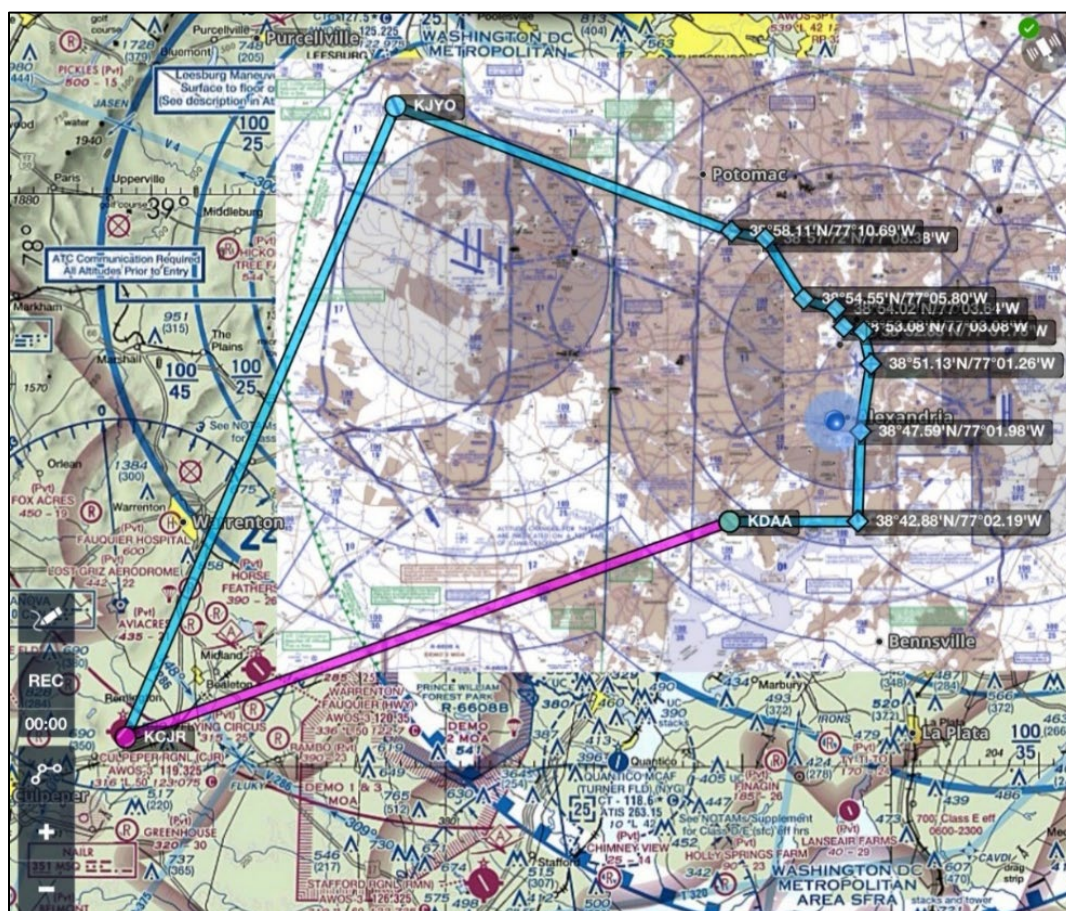


Figure 1. Planned route filed with DAA base operations as recreated on ForeFlight.

About 1600, the IP approached B Company's commander, who served as final mission approving authority (FMAA) for low-risk flights. The IP described his planned LZs and went over the completed R-COP. The commander told investigators that he had an abbreviated conversation with the IP about the mission, because they had already recently discussed relevant considerations in detail for previous NVG APART flights. He recalled reminding the IP of some things they had previously discussed,

⁸ Emergency Response Method (ERM): (F) fly the aircraft, (A) alert the crew, (D) diagnose the emergency, (E) execute emergency procedure, (C) communicate the plan, (F) fly the aircraft.

⁹ EP - Emergency procedures

¹⁰ 1070 is a task in the Army flight crew training manual involving response to emergencies.

¹¹ Interview with CW4 Matthew Ingalls, p. 20. The IP noted in the R-COP that night unaided flight will only be along helo routes when cultural lighting becomes a hazard to NVG operation.

such as ensuring that he did not become too fixated on his evaluator role and that he remained in control of the helicopter.¹² The commander approved the mission.

According to the mission packet, a crew brief was planned for 1600. B Company records indicated the IP signed out four sets of NVGs from a secure storage area in the B Company hangar (two white phosphor and two green phosphor NVGs) (Attachment 4). The IP obtained keys to the helicopter from the 12th Aviation Battalion flight operations building.¹³ The crew received a weather briefing at 1725. The IP texted his girlfriend at 1725 that he was "going flying".¹⁴ The CE texted his wife at 1741 that he was "taking off".¹⁵ The PI texted her boyfriend at 1750 that she was "about to take off".¹⁶ The flight plan filed with base operations indicated a planned departure time of 2320 UTC (1820 EST), and the 12th Aviation Battalion Duty Officer's Log noted the helicopter's takeoff time as 1842.

ATC mode C (altitude reporting) radar data files provided by the FAA indicated that returns from the helicopter first appeared on radar northwest of DAA at 1846 (Figure 2). The helicopter proceeded south along the east bank of the Potomac River, then turned west near Fredericksburg, VA. The helicopter proceeded to the Greenhouse Airfield (02VA), did an orbit, and landed. The helicopter departed 02VA about 1930 and proceeded north for about 52 nm. As it passed Hamilton, VA, it turned east towards the Federal Support Center (1MD5) where it orbited and landed. The helicopter departed 1MD5 about 2029 to the south at an altitude between 1,200 and 1,500 feet msl. The crew checked in with DCA tower at 2032 with a request to proceed to DAA via direct Cabin John, helicopter route 1, and transition to route 4. The controller issued an altimeter setting of 29.89 inHg which the crew correctly read back.

Radar data indicated that the helicopter arrived at Cabin John at 2038 and 1,400 ft msl (Figure 3). It turned west, performed a 180-degree left turn to the east, proceeded eastbound over the Potomac River, and joined helicopter route 1. The Army Aviation Brigade (TAAB) SOP required aircrews to fly no less than 100 ft below the maximum altitude for the helicopter route being flown to accommodate noise

¹² The company commander stated that when serving as an FMAA he normally focused on the role the instructor pilot plays, acting as the PI while they are actually the IP, decision making while evaluating required tasks, and making sure they take control when appropriate to ensure safety. He also emphasized maintaining airspace awareness and deconfliction while busy with other tasks. (Interview with CPT Phillip Gallon, p. 9)

¹³ Confirmed February 7, 2025, by in-person verbal communication between the group co-chairs and an army service member who was on duty in the operations center at the time and provided the keys.

¹⁴ Summary of interview with [REDACTED], p. 2.

¹⁵ Second interview with [REDACTED], p. 29.

¹⁶ Interview with the PI's boyfriend, p. 17.

abatement concerns.^{17,18} The published transit altitude between Cabin John and Chain Bridge was at or below 1,300 feet msl.

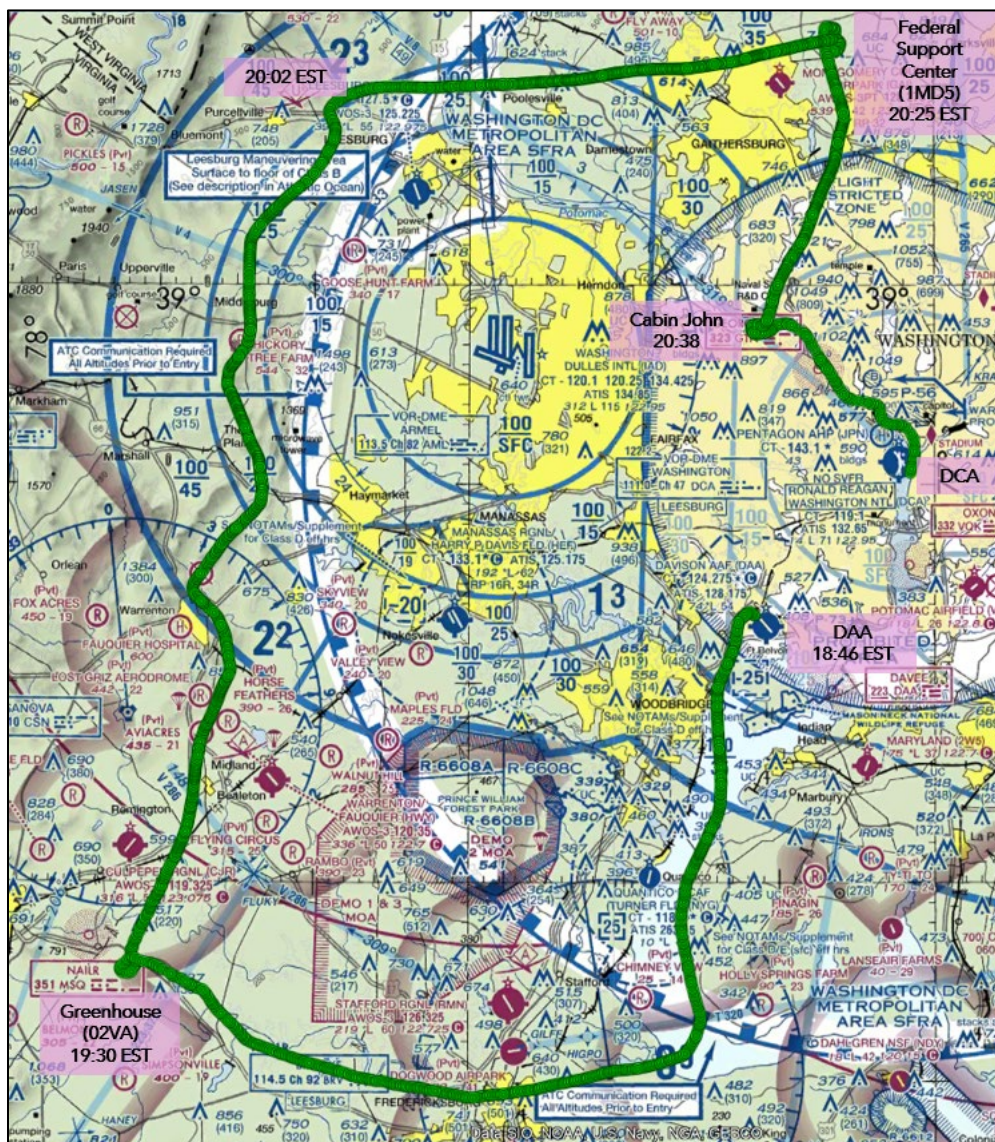


Figure 2. PAT 25 actual route of flight derived from radar data.

The helicopter crossed Chain Bridge at 2042 and Key Bridge at 2044. The published transit altitude between those bridges was at or below 700 feet msl. After Key Bridge the published transit altitude dropped to at or below 300 feet msl. The Blackhawk crew checked in with DCA Tower at 2045:14 and stated, "PAT 25 memorial", as required, before flying over the Arlington Memorial Bridge at 2045:30. After that, the published transit altitude was at or below 200 feet msl.

¹⁷ Interview CPT Sawatzky, Flight Operations Officer-in-Charge (OIC)

¹⁸ TAAB SOP, Sec 4-19

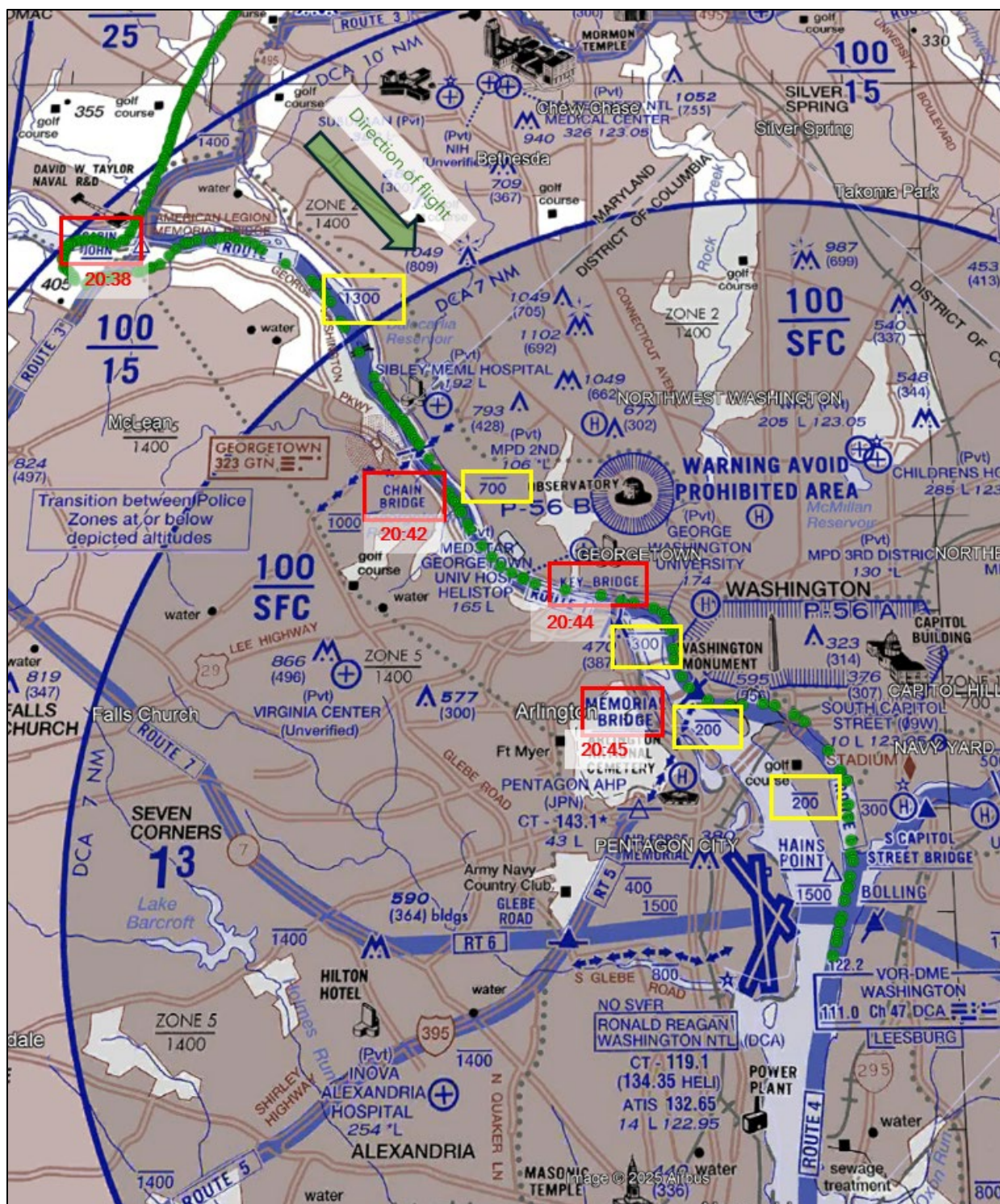


Figure 3. Helicopter Route 1 and Route 4 depicted on the Baltimore-Washington Helicopter Route Chart dated December 26, 2024. The Blackhawk helicopter radar track is depicted by green dots. VFR check points and times are highlighted by red boxes and the route step down altitudes highlighted in yellow boxes.

DCA Tower contacted the PAT 25 at 2046 and stated, "Traffic just south of Wilson Bridge is a CRJ at one thousand two hundred feet circling runway three three".¹⁹ The Blackhawk IP responded "PAT 25 has the traffic in sight request visual separation."²⁰ DCA Tower approved the request. The helicopter crossed Hains Point at 2047. At 2047:39, or 15 seconds before collision, DCA Tower asked, "PAT 25, you have the CRJ in sight?" At 2047:42 DCA tower transmitted, "PAT 25 pass behind CRJ." The Blackhawk CVR recording indicated that the phrase "pass behind" was stepped on by a 0.8 second microphone key press by the crew of the helicopter. The Blackhawk replied at 2047:44, "Aircraft in sight, request visual separation." The controller approved the request. The Blackhawk CVR indicated the crew discussed their location and what the controller wanted them to do. They decided to fly closer to the east side of the river. The collision occurred shortly after this exchange.

2.0 Personnel Information

2.1 The Instructor Pilot (IP)

The IP, age 39, was a CW2 assigned to B Company, 12th Aviation Battalion as a platoon instructor pilot. He joined the US Navy in 2007 and served as a nonflying aviation crewmember. He transitioned to the US Army in 2017 because he wanted to serve as a pilot. He was commissioned as a warrant officer in 2018. He completed initial flight training at Fort Rucker (now Fort Novosel) May 2, 2019. He was stationed at Fort Belvoir and assigned to A company, 12th Aviation Battalion, TAAB, as a pilot on June 3, 2019. He was assigned to A company, 1st BN, 228th Aviation Regiment, and deployed to Honduras September 25, 2023. He was assigned to B Company, 12th Aviation Battalion, TAAB, and returned to Fort Belvoir October 1, 2024.

A battalion standardization pilot considered the IP one of his best instructors. He thought the IP was very objective.²¹ A B Company pilot who had received instruction from the IP thought he was average compared to other instructors. B Company's safety officer had also received instruction from the IP and he described him as "by the book." The safety officer said the IP was not known as a particularly difficult instructor, but that he held pilots to the standards. The 12th Aviation Battalion commander had also flown with the IP before and considered him "very good."

2.1.1 Recent Activities

The IP was married. The IP and his wife separated in 2023 when he deployed to Honduras. Upon his return to the US in October 2024, the IP established a residence at a boarding house in Alexandria, Virginia. His wife and children lived in

¹⁹ The Wilson Bridge was about 5.5nm south of the helicopter's position.

²⁰ A full discussion of the aircrafts' visibility to one another will be discussed in the Visibility Study [].

²¹ Interview with CW4 Matthew Ingalls, p. 16.

Lorton, Virginia. His wife told investigators that she and the IP had been reconciling since his return.

The IP had a girlfriend who resided in Sacramento, California. Cell phone records indicated that the IP and his girlfriend communicated frequently by phone calls and text messages. The IP's girlfriend said they had been getting along well, their recent communications had been casual and routine, and they planned to move in together in two years.

The IP's wife said the IP did not mention feeling stressed, tired or napping in the days before the accident. The IP's girlfriend said she thought his family situation and living arrangements likely caused some stress, but she thought he seemed okay.

The IP's wife estimated that the IP needed seven or eight hours of sleep per night to feel fully rested. Information about the IP's recent activities was obtained from interviews and cell phone records.²²

Sunday, January 26, the IP was off duty. His cell phone records showed extensive call activity between 1341 and 2210, mostly with his girlfriend. Neither his wife nor his landlord recalled anything about his activities. The IP's last cell phone activity was an outgoing call to his girlfriend that began at 2234 that ended at 2236.

Monday, January 27, the IP was on duty from 0530 to 1700. He was paired with the 12th Aviation Battalion's safety officer. They prepared a helicopter for flight and spent the rest of the day waiting nearby. The IP's first cell phone activity was a 14-second inbound call at 0717. The IP spent that evening in Lorton with his kids. His wife contacted him that evening to let him know she was not feeling well, and he picked her up from work and brought her back to her residence. The IP's wife told investigators that upon arriving she went to sleep. She said the IP likely slept at their home, but she did not know when he slept. His last cell phone activity was a call from his girlfriend that began at 2158 and ended at 2342.

Tuesday, January 28, the IP's wife left for work at 0600. She did not know when the IP awoke. The IP had an office day at DAA, which typically involved an 0900 to 1700 schedule. His first cell phone activity was a 1-minute inbound call at 0804. He sent texts and participated in other calls between 1017 and 1230. He texted his wife at 1230 about a home they were considering purchasing together. At 1900, he responded immediately to a text from his landlord asking him to turn off the oven, which indicated to his landlord that the IP was at their house in Alexandria. The IP and his landlord later watched television together and the IP retired to his room about 2100. Neither the IP's landlord nor his wife knew when the IP went to sleep. Cell phone records showed extensive call activity on the IP's phone between 1733 and

²² See interviews with [REDACTED].
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2135. The last activity was a call to his girlfriend that began at 2323 and ended at 0005 (January 29). The IP's girlfriend said this call was routine and positive, and she believed he went to sleep immediately afterward.

On Wednesday, January 29, the IP went to work later than usual because of his planned flight with the PI that evening.²³ The first activity on his cell phone was an outgoing text at 0932. His landlord said a friend interacted with the IP at the boarding house as the IP was making breakfast. The IP explained to the friend that he was going to work later than usual because he had an evening flight. The IP texted his wife about routine matters at 1252. The MBO who briefed the IP for the accident flight recalled that the IP appeared normal. The company commander who approved the mission told investigators the IP seemed "very alert."²⁴

2.1.2 FAA Certifications

The FAA indicated that the IP had the following pilot certifications and ratings.

May 10, 2019	Commercial pilot - rotorcraft helicopter Instrument helicopter S-70 Type rating
--------------	---

2.1.3 Army Authorized Flight Duties

US Army records (Attachment 5) indicated that the IP had the following flight authorizations designated by the Aircrew Training Program (ATP) commander (captain of B Company):

- UH-60L
 - Instructor pilot (IP), Pilot-in-command (PC), right and left seats.
 - Co-pilot (CP) and Instructor pilot (IP), cabin.
 - Co-pilot (CP) Pilot (PI), pilot in command (PC), instructor pilot (IP), NVG.

2.1.4 Flight Experience

US Army records (Attachment 5) indicated that the IP had the flight experience summarized in Table 1.

Table 1. The IP's flight experience.

Total Flight Time	968.2
-------------------	-------

²³ 12th Aviation Battalion personnel told investigators they normally plan to start their duty day 12 hours before they expect their flight activity to end to avoid exceeding a maximum duty day.

²⁴ Interview with CPT Phillip Gallon, p. 17.

Last 60 days	25.5 hours, 12 flight (12.7 hours and 4 flights were NVG)
UH-72A	82.0 hours
UH-60M	584.7 hours
UH-60L	301.5 hours
Pilot in Command (UH-60)	295.6 hours
Total Simulator	109.4 hours
Instructor (UH-60L)	157.7 hours

2.1.5 Training History

The IP's aviation training history, obtained from US Army records (Attachment 5), is summarized in table 2.

Table 2. The IP's aviation training history

May 2, 2019	Completed US Army helicopter flight school
June 12, 2019	FAA records show he completed DC Special Flight Rules Area (SFRA) online training.
September 9, 2019	Completed training readiness level 3 (RL3) training, designated readiness level 2 (RL2)
September 23, 2019	Completed RL2, designated readiness level 1 (RL1)
October 15, 2021	First PC designation on the UH-60
August 30, 2023	Completed IP Course
October 16, 2023	Completed UH-60M to UH-60L training
November 02, 2023	Successfully completed first PC evaluation on the L Model
January 24, 2024	First IP evaluation SAT
November 12, 2024	Completed annual Aircrew Coordination Training (ACT)
November 14, 2024	Instrument and Standardization Day / NVG evaluation SAT

2.1.6 Recent Flight Activity

The 12th Aviation Battalion provided flight records for the IP. Investigators tabulated actual and simulated flight hours the IP logged each month for a year preceding the accident (figure 4), excepting the accident flight. The IP had flown 269 hours in the last year, an average of 22.4 hours per month. He had flown 98.7 hours in the last six months, an average of 16.5 hours per month. In the month before the accident, the IP flew on January 15 (5.0 flight hours) and January 16 (4.0 flight hours) (Attachment 5). Both were training flights. He was scheduled to fly but did not fly on January 8, 9, and 22 (Attachment 5).

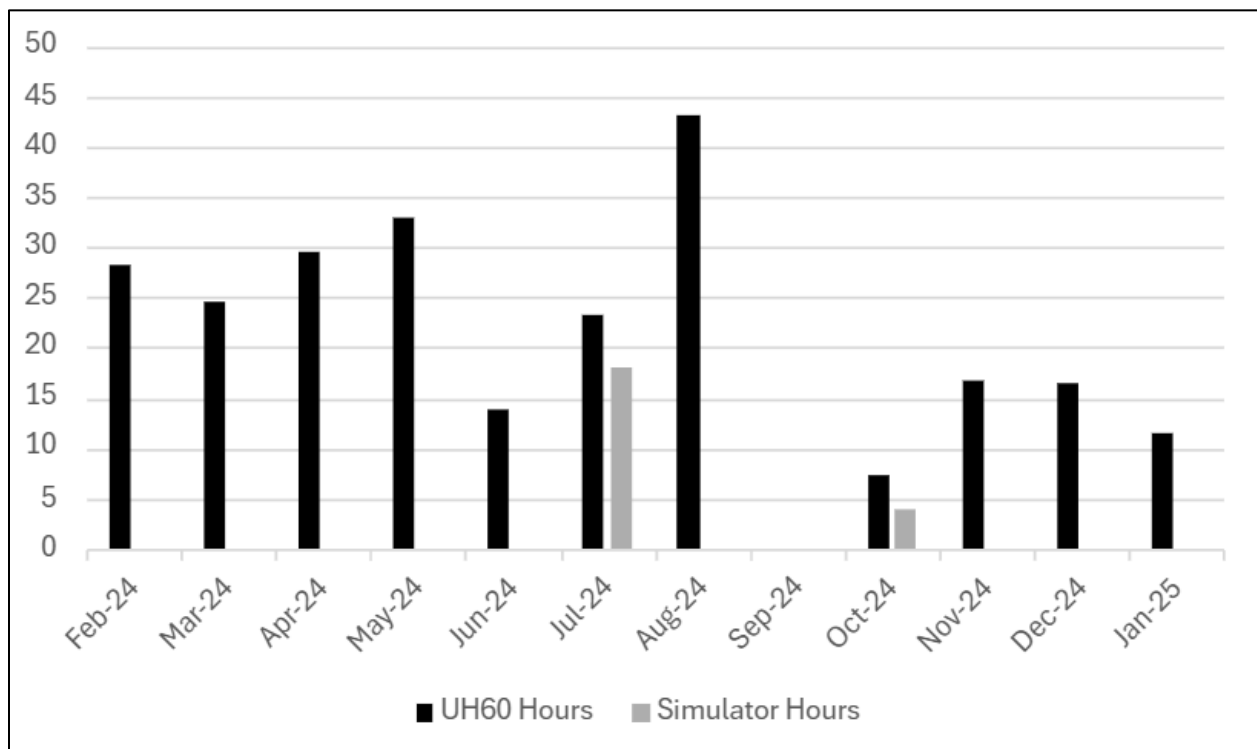


Figure 4. The IP's UH60 actual and simulated flight hours, February 2024 to January 2025.

2.2 The Pilot (PI)

The PI, age 28, was a Captain assigned to Headquarters Company, 12th Aviation Battalion, TAAB, Military District Washington (MDW), as a staff officer. She had served as a ROTC cadet in college, and she had been commissioned as a second lieutenant on November 3, 2019. She completed flight school at Ft. Rucker (Ft. Novosel) July 9, 2021. She was stationed at Fort Belvoir July 19, 2021, and assigned as platoon leader for B Company, 12th Aviation Battalion on July 1, 2022. On April 10, 2023, she was named B Company's executive officer. December 5, 2023, she was

named assistant operations officer (S3) for the 12th Aviation Battalion, but she remained with B Company's aircrew training program (ATP) for flying purposes. The PI was subsequently assigned as platoon leader for a new petroleum oils and lubricants platoon on August 19, 2024, an assignment she maintained until the time of the accident.

A brigade standardization pilot who provided some training to the PI when she arrived at the 12th Aviation Battalion in 2021 said the PI was eager to learn but initially had weak flying skills. Her situational awareness and leadership skills were above average for her experience level but her "stick and rudder skills" were below average. He said the L model was "particularly taxing to fly to maintain altitude and air speed" while "understanding and utilizing the trim system" for students (like the PI) who initially trained on the more automated UH-60M model. He recalled working extensively with the PI on how to properly use the trim system and the collective friction lock when flying on instruments and he signed her off as fully qualified RL1 (readiness level 1) in the UH-60L on January 6, 2022. He subsequently conducted evaluation flights with her in January and May 2023 and January 2024, and he estimated that he had flown with her a total of 20-40 times. His overall impression was that her aptitude as a pilot was average.

A battalion standardization pilot who conducted an instrument evaluation with the PI on February 17, 2022, found her performance "well below average". He described her as one of the bottom three pilots he had trained who had come directly from primary flight school. He described her as having "lots of difficulties in the aircraft." He said she got fixated on engineering details of the helicopter that were irrelevant to the tasks at hand, and he had to redirect her toward mastering flying skills. He reverted her to the lowest readiness level and trained her extensively for about a month until she demonstrated full readiness on March 16, 2022. Asked why he thought the PI had been designated full readiness by the previous standardization pilot, he said:

I think it was several things. You know, the difficulty of showing up to a unit you've got very little helicopter time, and you have absolutely no Lima [UH-60L] time, so now you're, you're downgrading, in a way, from an advanced aircraft to an aircraft that [has] steam gauges. You're being evaluated and trained by a new instructor pilot which is fresh out of the instructor pilot course, so he's still developing his teaching methods and instructing methods and, you know, all those things kind of, you know, came probably -- came together with one another to, you know, not help the situation as far as her instruction goes.²⁵

²⁵ Interview with Brad Haase, p. 21.

B Company's commander recalled flying with the PI twice in 2023. He said that the PI knew the mission well and "was a highly achieving, high motivated and driven aviator who was actually capable of accomplishing a lot more than I've seen some of the more junior pilots of her hour level who were similar to her, being able to accomplish in some of the combat aviation brigades."²⁶ He added that she was "great on controls, great at process management and workload management in the aircraft." His overall impression was that she was "a solid pilot-in-command who was capable of executing and had that knowledge, had glimmers of advanced knowledge for where she was."²⁷ A 12th Aviation Battalion flight operations officer in charge recalled flying with the PI on one occasion before she qualified as a pilot in command (which occurred on January 11, 2024). He recalled that the flight went "fine."

A brigade standardization pilot who conducted an instrument evaluation with the PI in January 2024 and had only flown with her that one time described her as "incredibly professional", diligent, and thorough. He said her performance as a pilot was "always good" and she had a good reputation as a pilot and officer.²⁸ A C company standardization pilot who had flown with the PI twice said she was "well organized" and "an incredible planner". He said, "I don't think I've seen any more professional officer in this unit or really any unit that I've been in." He could not recall anything specific about a successful proficiency flight evaluation (PFE) he conducted with her in August 2024, which led him to believe that her performance was average.

A B Company pilot who flew with the PI in September 2024 recalled that the PI appeared deeply knowledgeable. He said that her flying was "nothing out of the ordinary." The 12th Aviation Battalion's commander had flown with the PI on two occasions, most recently in October 2024, said the PI was detail-oriented and she said that she had no issues with the PI's flying. B company's safety officer had performed one flight with the PI in December 2024. He recalled that the PI arrived prepared, had a plan, and knew exactly what she was doing. He was "very impressed" by her organizational skills and her attitude towards flying and being in the Army.

2.2.1 Recent Activities

The PI shared an apartment with a roommate in Washington, DC. She had a boyfriend who did not live with her. Her boyfriend recalled that the PI had seemed happy in recent days. He estimated that she needed six to eight hours of sleep per night to feel rested. The PI's roommate said the PI had difficulty consistently obtaining eight hours of sleep at night, and she seemed "quite happy" when she got six.

²⁶ Interview with CPT Phillip Gallon, p. 12.

²⁷ Interview with CPT Phillip Gallon, p. 23.

²⁸ Interview with CW4 Matthew Ingalls, p. 17.

Information about the PI's recent activities was gleaned from witness interviews and cell phone records.²⁹

On Sunday January 26, the PI attended a dance class with her boyfriend from 1600 to 1700. Her sister, who resided in Florida, said the PI participated in a family video call from 1817 to 1830 and appeared cheerful. The PI and her sister exchanged text messages between 1844 and 2147. The PI was home watching television. The PI called and texted her boyfriend between 2210 and 0025. He recalled that these communications were routine, and he thought she likely went to sleep afterward.

Monday, January 27, the PI texted her boyfriend upon waking at 0843. They spoke by phone at 0905. The PI texted her roommate about a routine matter at 1015. She had a flight that day with a 1200 showtime. The PI texted her boyfriend that she departed at 1709 and landed at 1936. A B Company pilot who flew with the PI said she practiced basic flight maneuvers and NVG tasks near DAA and NYG in preparation for her upcoming proficiency check. He recalled that the PI seemed "rusty" on the controls due to a lack of recent flying, but otherwise nothing stood out about her performance. The PI texted her sister at 2114 and her roommate at 2200 about routine matters. The PI's boyfriend arrived at her apartment about 2300. He fell asleep there between 2330 and 2400. He did not know when the PI fell asleep. Her last cell phone activity (a text message to her sister) was at 0115.

Tuesday, January 28, the PI and her boyfriend woke at 0630. Her boyfriend left her apartment about 0700. When he left, the PI told him she planned to go back to sleep, however cell phone records indicated she received an incoming call at 0734 and it lasted 52 minutes. The PI texted her boyfriend a greeting at 0850. She did not have a flight that day, and her office schedule was 0900 to 1700. Between 1058 and 1317 She texted her roommate, boyfriend, and sister about having toured a house for sale that morning. A Bravo Company standardization pilot recalled seeing the PI in the pilot planning room about 1800. They exchanged greetings. She appeared fine to him and did not express any concerns. The B company commander also saw the PI in the planning room that evening and recalled that she looked focused. A B Company pilot also saw her in the planning room and shared tips with her about their flight planning software and some landing zone SOPs.³⁰ Between 1900 and 2000, the PI participated in a call with a nonprofit that helped military personnel transition from military service to academic study. At 2142, the PI texted her boyfriend and said she was "still working." Her last cell phone activity was a text exchange with him at 2325 about routine matters. She did not indicate to her boyfriend when she intended to go to sleep, but she told him she intended to wake at 0700.

²⁹ See interviews with [REDACTED].

³⁰ Interview with Adrian Stowers, pp. 8, 10.

Wednesday, January 29, the PI's boyfriend called her at 0707 to make sure she was awake. They spoke for about ten minutes. He recalled that she sounded happy.³¹ He texted her at 0730, concerned about high wind in the forecast, and she responded that the forecast was okay. According to her boyfriend, the PI attended physical therapy near her residence at 0900. Between 1137 and 1206, the PI texted her roommate, boyfriend, and sister that she was making an offer on the house she had toured the day before. At 1218 she had a 4-minute call with her boyfriend about routine matters and she told him she had to handle something with her fuel platoon before leaving for her evaluation flight. Showtime for the evaluation flight was 1300. The captain in charge of Bravo company recalled seeing the PI in the planning room and he recalled that she appeared focused. The B Company pilot who had shared tips with the PI the day before interacted with her again. He recalled that she seemed excited and not apprehensive about her proficiency check. The B Company pilot who had flown with the PI on January 27 saw her getting ready to depart the B company offices for the flight line. He recalled that she appeared fine.

2.2.2 FAA Certifications

The PI's FAA pilot certifications and ratings, obtained from FAA records, are listed below.

July 15, 2021	Commercial pilot - Rotorcraft helicopter Instrument helicopter S-70 Type rating
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2.2.3 Army Authorized Flight Duties

The PI had the following flight authorizations designated by the Aircrew Training Program (ATP) commander (the captain of B Company) (Attachment 6):

- UH-60L
 - Pilot (PI), Pilot in Command (PC), left and right seats.
 - Co-pilot (CP), cabin
 - Co-pilot (CP), Pilot (PI), pilot in command (PC), NVG

2.2.4 Flight Experience

The PI's flight experience, obtained from US Army records, is summarized in table 3 (Attachment 6).

³¹ He told investigators the PI had not been feeling nervous about her evaluation.
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Table 3. The PI's flight experience.

Total Flight Time	454.3 hours
Last 60 days	4.4 hours, 5 flights (2.9 hours NVG)
UH-72A	83.6 hours
UH-60M	44.5 hours
UH-60L	326.2 hours
NVG	136.5 hours
Pilot in Command (UH-60)	44.67 hours
Total Simulator	86.5 hours

2.2.5 Training History

The PI's aviation training history, obtained from US Army records, is summarized in table 4 (Attachment 6).

Table 4. The PI's aviation training history.

July 9, 2021	Completed US Army helicopter flight school
July 30, 2021	Commander's evaluation, unit training started, designated RL3
August 24, 2021	DC SFRA training completed (FAA record)
October 29, 2021	Training record note: "ACM ³² has had their RL Progression NLT date adjusted due to (commander approved leave and cancellations due to wx). New progression date is 15NOV2021."
November 16, 2021	Training record note: "RL progression timeline adjusted. ACM has had their RL progression NLT date adjusted due to canceled flight 9NOV2021. New progression date 16NOV2021."
November 17, 2021	Training note: "ACM failed to complete the following ATP requirements. Progress from RL3 to RL2 within 90 days. This was due to ACM conducting UH-60M to L transition in

³² ACM - Aircrew Member
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	conjunction with RL3 progression and reduced availability of instructor pilots."
November 17, 2021	"IAW TC 3-04.11 [Training Circular] When an ACM has not progressed to the next RL within the time specified the unit commander will take action according to AR95-1. In accordance with AR95-1 paragraph 4-10 CDR investigation initiated."
November 22, 2021	"Request for extension to ATP requirements. ACM requests 30-day extension for failure to meet ATP requirements IAW AR95-1 para 4-10 [Army Regulation], 30-day extension will be utilized to address task 1166 (perform instrument maneuvers) and task 1174 (perform holding procedures) IAW UH-60 DATM. ACM will fly or attend simulator 2-3 times a week in order to attain proficiency in tasks listed. ACM must attain RL2 NLT December 22, 2021."
November 29, 2021	Completed RL3 training, extension requirements completed
January 6, 2022	Completed RL2 training, designated RL1
February 17, 2022	Failed NVG annual evaluation. The following tasks were failed: • 1048 "No fuel consumption checks started or closed out over two hour flight" • 1052 "ACM could not maintain appropriate common standards altitude deviations of 200 ft and airspeed of up to 20 KIAS and greater than full ball width of trim" • 1058 "ACM could not maintain a constant angle for approach or ground track alignment with landing direction" • 1070 "(Could not evaluate three EPs due to aircraft time limitations, ran out of time trying to get appropriate common standards within standard and runway alignment for landing from the left seat) Tasks 1068, 1064, 1062, 1184 still need to be evaluated. Recommend RL3 D/N/NG and ACM conduct retrain and evaluation of annual NG stands evaluation, IAW TC 3-04.11 para 8-48, ACM has 90 days to complete the required training."

February 28, 2022	"ATP requirement not complete. ACM failed to complete the following ATP requirements: Second period hours 0.5 H/W [hood/weather], Annual 0.8 sim, standardization & NVG flight evaluation. This was due to removal from RL due to training deficiency. ACM has 90 days from 17FEB22 18MAY22 to make RL1. ACM will complete all requirements prior to designation to RL1."
March 1, 2022	Commander investigation initiated due to ACM not completing ATP requirements. Commander granted 30-day extension.
March 16, 2022	ACM completed NVG annual evaluation. APART complete. Redesignated RL1.
August 31, 2022	ACM completed DC SFRA training (FAA record)
January 31, 2023	Instrument evaluation - SAT
February 6, 2023	NVG annual evaluation - SAT
March 1, 2023	Temporary medical suspension from flight duty
May 15, 2023	End of medical suspension, ACM completed performance flight evaluation (PFE)
January 8, 2024	Instrument evaluation - SAT
January 11, 2024	NVG annual evaluation - SAT Initial designation as PC
January 26, 2024	Annual Aircrew Coordination Training (ACT)
March 16, 2024	Temporary medical suspension from flight duty
August 6, 2024	Termination of temporary medical suspension
August 15, 2024	Return to flying performance flight evaluation (PFE) to reestablish aircraft and NVG currency
November 15, 2024	Air mission commander academic training

2.2.6 Recent Flight Activity

The 12th Aviation Battalion provided flight records for the PI (Attachment 6). Investigators tabulated the number of actual and simulated flight hours the PI logged each month for a year preceding the accident (figure 5), not including the accident flight. The PI had flown 56.7 hours in the last year, an average of 4.7 hours per month. The PI had flown 39.1 hours in the last six months, an average of 6.5 hours per month.

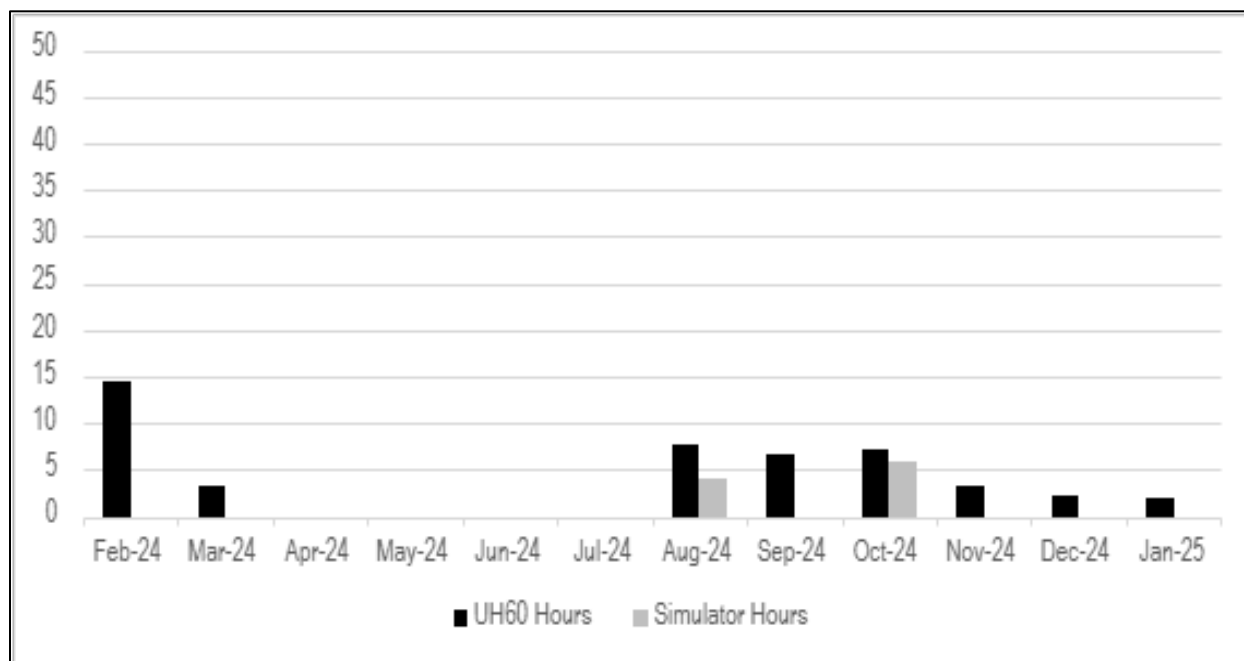


Figure 5. The PI's UH-60L actual and simulated flight hours, February 2024 - January 2025.

The PI's most recent flights in the month before the accident were training flights conducted on January 22 (0.2 hours) and January 27 (1.9 hours). The PI had one canceled flight in the month of January, on January 22 (Attachment 6).

2.3 The Crew Chief (CE)

The CE, age 28, was a staff sergeant (SSG) assigned to B Company, 12th Aviation Battalion. He joined the US Army July 14, 2014. He completed basic combat training at Fort Jackson, SC and advanced individual training as a UH-60 repairer at Fort Eustis, VA. His first duty assignment was with D company, 2-158th Aviation Battalion, Joint Base Lewis-McChord, Washington. In 2016, he was assigned to A company, 2-158th Aviation Battalion where he served as a crew chief and helicopter repairer and deployed to Afghanistan. In 2017, he was assigned to the 12th Aviation Battalion, Fort Belvoir, VA. In 2020, he was reassigned to C company, 2nd Battalion, 3rd Combat Aviation Brigade, Hunter Army Airfield, Georgia, and deployed to Europe. January 1, 2023, he returned to the 12th Aviation Battalion assigned to A

company, and on January 3, 2023, he was assigned to B Company as a helicopter repairer and crew chief. Since that time, he had picked up an additional responsibility as an enlisted instructor (FI).

A battalion standardization pilot told investigators he had recently completed the CE's annual performance evaluation, and he considered the CE one of the best and most intelligent instructors he had worked with. The CE continued to perform maintenance on the helicopters but he was also always teaching somebody something, developing a class, coming up with a training plan, or attending a weekly flight schedule meeting.³³ He stated that the CE was very good at spotting and calling out traffic using the battalion's standard terminology and that the CE provided aircrew coordination recurrent training to other CE's.

2.3.1 Recent Activities

The CE was married and lived with his wife and baby in Alexandria, Virginia. His wife said he had experienced mental health symptoms resulting from work related burnout in 2024 that had prompted him to remove himself from flight duty. She said that he was subsequently authorized to return to flight duty. She reported that in recent weeks he had experienced frustration balancing work and home responsibilities but seemed okay. She estimated that he needed 7 to 9 hours of sleep per night to feel fully rested.

B Company's sergeant first class (SFC) was a close colleague and friend of the CE. He told investigators that the CE and his wife were separating after a long period of relationship difficulties. The SFC said that on the day of the accident he and the CE made plans to meet up the following day so the SFC could help the CE figure out his future living and childcare arrangements. The SFC said the CE seemed clear-headed and at peace with the decision to separate from his wife and that he had expressed positive feelings about his future.³⁴

Information about the CE's recent activities was gleaned from interviews and cell phone records.³⁵ On January 26, 2025, the CE was off duty. His wife could not recall anything about his activities or when he went to sleep. His last cell phone activity was a text message he sent at 2259.

January 27, 2025, the CE's wife recalled that the family woke about 0700. The CE's first cell phone activity was a text message sent at 0901. The CE and his wife attended a family counseling appointment at 1000 and the CE's wife recalled that it went well. The CE left home for work about 1200. The last activity on his cell phone was an outgoing text at 1255. The CE's wife estimated that he returned home from

³³ Interview with CW4 Matthew Ingalls, p. 18.

³⁴ Interview with SFC Wayne Bulson, p. 8.

³⁵ See Interviews with [REDACTED], Wayne Bulson, William Barton.

work between 1700 and 1730. She recalled that he was home for dinner and they went to sleep about 2130.

January 28, 2025, the CE's wife could not recall when the CE woke or left the house, only that he had left early. His first cell phone activity was a five-minute outgoing call at 0923. His wife recalled that he came home about 1100 complaining of a sinus headache and took a nap for two or three hours. His cell phone records showed outgoing text messages at 1114 and 1131, a six-minute incoming call at 1233, and a 90 second outgoing call at 1531. There were no additional texts or calls on his phone until 1942. The CE told his wife he felt better after the nap. They engaged in routine activities at home for the rest of the evening. His wife did not know when he went to sleep. She went to sleep before him about 2100-2130. The last activity on the CE's cell phone was an outgoing text at 2244.

January 29th, 2025, the CE's wife could not recall when the CE awoke. She told investigators that she left the house with their child at 0730 to visit a nearby park and avoid waking the CE. The CE's cell phone records showed numerous text messages between 0603 and 0802 and phone calls made and received at 0919 and 0938, respectively. His wife recalled returning from the park about 1030 or 1045. The CE was in the kitchen washing dishes. He seemed "like his normal self." He did not seem tired. As she entered their residence, she recalled that her knee bumped a door frame and began to swell. About 1300 a friend drove her to an urgent care clinic and the CE drove their son to a babysitter. The CE and his wife were concerned about whether his wife's injury would prevent her from driving their vehicles because they had manual transmissions. The CE texted his wife at 1345 indicating that he had arrived at the B Company hangar. His wife texted him at 1604 to let him know that her knee would be okay. She texted him again at 1612 to tell him she was going to try driving her car. She texted him at 1638 that she had been able to drive it. She told investigators that the CE seemed to have a "low, normal" level of concern about her injury, and the event did not seem a "significantly stressful" for him.

The B Company SFC saw the CE arrive between 1300 and 1400 and begin examining the helicopter's maintenance logbook. The SFC said this was the CE's routine before every flight. They had a brief conversation, and the CE told him he would talk with him later. The SFC recalled that the CE seemed very focused on getting ready for the flight. B Company's safety officer recalled seeing the CE and the IP in the commander's hallway about 1600. He recalled that the CE looked "normal". The CE made a brief call at 1634 and received a brief call at 1636. B Company's first sergeant saw the CE and PI preparing to go to the helicopter just before 1730. He told investigators the CE looked "perfectly fine". The CE's wife texted him at 1943 to ask if he had landed yet. At 2024 he replied that he had not. At 2025, she texted him to say she was sorry because it was a long flight. At 2026, he responded, "It is what it is." That was the last communication she received from him.

2.3.2 FAA Certifications

The CE's FAA pilot certifications and ratings, obtained from FAA records, are listed below:

June 29, 2023 Airframe & Powerplant Mechanic

2.3.3 Army Authorized Flight Duties

The CE had the following US Army flight authorizations designated by the Aircrew Training Program commander (the captain of B Company) (Attachment 7):

- UH-60L
 - Crew chief (CE), Crewmember flight instructor (FI), cabin
 - Crew Chief (CE), Crewmember flight instructor (FI), NVG

2.3.4 Flight Experience

The CE's flight experience, obtained from US Army records (Attachment 7), is summarized in table 5.

Table 5. The CE's flight experience.

Total Flight Time	1149.1 hours
Last 60 days	14.3 hours, 9 flights (7.0 NVG hours)
UH-60M	448.5 hours
UH-60A	4.8 hours
UH-60L	185.9 hours
HH-60M	486.7 hours
NVG	392.8 hours
Instructor	4.0 hours

2.3.5 Training History

The CE's aviation training history, obtained from US Army records (Attachment 7), is summarized in table 6.

Table 6. The CE's aviation training history.

January 23, 2015	Awarded aviation badge - basic following completion of aviation MOS training.
February 11, 2015	UH-60M helicopter repairer course completed
September 23, 2015	UH-60L helicopter repairer course completed
August 16, 2016	Completed nonrated crewmember qualification training
December 14, 2017	Completed basic leader's course
February 14, 2019	Completed aircraft crewmember standard instructor course
May 31, 2024	Completed advanced leader's course
July 9, 2024	Annual Aircrew Coordination Training (ACT)
June 18, 2024	Completed standardization flight evaluation to become a CE standardization instructor
July 31, 2024	Completed APART evaluation
December 17, 2024	PFE ³⁶ flight - Reestablished NVG and aircraft currency

The CE's previous flights in the 30 days before the accident were training flights conducted on January 14 (4.0 hours) and 16 (4.0 hours).

3.0 Helicopter Information

An exemplar UH-60L from the 12th Battalion is shown in figure 6. The airframe of the UH-60L helicopter is divided into six sections: cockpit, cabin, transition, tail cone (tail boom), tail rotor pylon (vertical fin), and main rotor pylon. For the purposes of these field notes, the "main fuselage" is composed of the cockpit, cabin, main rotor pylon, and transition section. The primary structure is constructed of aluminum. Titanium and steel are used for firewalls and various fittings. Nonstructural members are primarily constructed of reinforced plastic. The non-retractable landing gear system is a reverse-tricycle configuration, with the two main landing gear (MLG) struts attached to the cockpit-cabin area and a tail wheel strut connected to the tail cone

³⁶ Proficiency Flight Evaluation (PFE) required if crewmember flight currency requirements have expired.

and tail rotor pylon. The external stores support system (ESSS) are two wing-like structures that are installed on the upper cabin frame of the helicopter and are each supported by two struts that attached to the lower cabin frame. Each ESSS wing has two sponsons, one inboard and one outboard. Items, such as external fuel tanks or weapons, can be attached to the sponsons.



Figure 6. A 12th Aviation Battalion UH-60L equipped with the crashworthy external fuel system (CEFS)

The helicopter operations and human performance working group examined a 12th Aviation Battalion's UH-60L, tail number 259, to become familiarized with crew stations, controls, displays, and the external field of view from crew stations. Photos from this examination are presented in figures 7 through 19.



Figure 7. Exterior view of exemplar UH-60L, tail number 259.



Figure 8. Side view of exemplar UH-60L cockpit exterior.



Figure 9. Exemplar UH-60L cyclic and collective.



Figure 10. Partial view of UH-60L instrument panel and center pedestal.



Figure 11. Right side instrument panel.



Figure 12. Close-up of radar altimeter, barometric altimeter, and airspeed indicator.



Figure 13. Exterior view of right crew chief station with window open.



Figure 14. Interior view of left crew chief station with window closed.



Figure 15. View of cockpit from left crew chief station.



Figure 16. External view from left crew chief's station with window open.



Figure 17. Forward view when leaning out of left crew chief window.



Figure 18. Aft view when leaning out of left crew chief window.



Figure 19. External view from left crew chief station with the window closed.

3.1 Transponder

The Blackhawk was equipped with a AN-123(V) Common Identification Friend or Foe (IFF) Digital Transponder that provides automatic radar identification, altitude, and surveillance reporting, which incorporates Automatic Dependent Surveillance Broadcast (ADS-B) technology (Figure 20). The AN-123(V) provides the following functions:

- Mode 1 - 4 digit octal mission code
- Mode 2 - 4 digit octal code aircraft identifier
- Mode 3/A - Combines military identification (Mode 3) with civilian secondary radar surveillance radar Mode A.
- Mode 4 - IFF or crypto secure cooperative combat identification

Mode 5 - Enhanced encryption of Mode 4

Mode C - Altitude reporting to ATC

Mode S - Provides data message elements to Mode S capable radars, such as airspeed, altitude, and location, which can include ADS-B compatible data.

Table 7. Transponder control functions³⁷

Control/Indicator	Function
Normal	The transponder is in the normal operating state with normal transponder functions enabled in the selected modes
TA	The transponder is in the normal operating state with Mode S and [Traffic Collision Avoidance System] TCAS enabled for Traffic Advisories (TA) only. Mode S is automatically enabled if previously disabled
TA/RA	The transponder is in the normal operating state with Mode S and TCAS enabled for both TA and resolution advisories (RA). Mode S is automatically enabled if previously disabled.

TCAS was not installed in the helicopter, hence the TA and TA/RA functions were inoperative.



Figure 20. AN-123(V) Transponder control head as installed in the Blackhawk.

³⁷ TM 1-1520-237-1UH-60A/L Operator's Manual
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Air traffic information provided by the FAA to the NTSB revealed that the helicopter's transponder was responding to Mode 3/A and Mode C interrogations but it was not transmitting ADS-B Out (Mode S).³⁸

3.2 Army ADS-B Requirements

Most military aircraft are equipped with transponders that provide ADS-B Out capability, however, an FAA-DoD MOA (Federal Aviation Administration - Department of Defense Memorandum of Agreement), signed May 10, 2024, stated that the Secretary of Transportation could not require military aircraft to install ADS-B equipment on special mission aircraft, and that DoD was to maintain a list of airframes that the DoD would not equip with ADS-B Out due to imminent retirement or mission requirements that did not align with the broadcast of ADS-B Out (e.g., when there were operational security (OPSEC) risks). The MOA was to be reviewed triennially (Attachment 8). US Code of Federal Regulations Title 14, Part 91.225 allowed operation of military aircraft without ADS-B.

TAAB SOP. Section 4-3 through 4-7, Emission Control, provided guidance to aircrews regarding use of transponder functions (Attachment 8).

"4-3 Aircrews conducting sensitive or classified operation shall consider flight identification emission during pre-flight planning.

4-4 Sensitive and /or classified missions include but are not limited to flights conducted ISO³⁹:

- Simulated and actual named contingency lines.
- Planned Joint Task Force (JTF)- National Capital Region (NCR) exercises.
- DV/VIP operations in which the itinerary is known/specified to be known or public knowledge.
- LAO⁴⁰ and training conducted to mission sites and/or utilizing mission routing.

4-5 Aircrews will emit Mode 3 A/C in lieu of ADS-B out.⁴¹

4-6 Aircrews will not switch transponder modes during the flight

4-7 Aircrews will not request ADS-B out from ATC, nor will they inform ATC that command and approved ADS-B Out operations."

³⁸ For details about air traffic data see the Air Traffic Control Group Chairman's Factual Report.

³⁹ In support of (ISO)

⁴⁰ Local Area Orientation (LAO)

⁴¹ The Mode S function is what can provide ADS-B out transmissions.

3.3 Electronic Flight Bag (EFB) & Stratus Portable ADS-B Receiver

The 12th Aviation Battalion used Apple iPad mini devices as the primary source of flight publications through the ForeFlight app service, commonly referred to as an Electronic Flight Bag (EFB). Both the IP and PI obtained iPad devices from the B Company flight planning room and these devices were recovered from the wreckage area but could not be downloaded.

ForeFlight had the capability of displaying FAA aeronautical charts, plot flight planned routes, (notices to airmen) NOTAMs, and weather information. The ForeFlight app could also display airborne traffic if those aircraft were broadcasting ADS-B data. Traffic targets could be overlaid on the aeronautical chart that was currently being displayed. Crews obtained portable ADS-B receivers (Stratus 2S devices) from the B Company flight planning room. These devices provided ADS-B In traffic information to the ForeFlight app on crew iPads via a Wi-Fi connection.

Stratus 2S was a portable wireless receiver made by Appareo that provided subscription-free access to electronic weather information, GPS information, backup attitude, and ADS-B traffic information. It was designed with an Open ADS-B Mode to work with a variety of popular apps – both iOS and Android.⁴² In response to an investigator inquiry, B Company provided a written statement indicating that the missing Stratus S/N: EO19251 was likely onboard, because all remaining Stratus devices in the Battalion were accounted for. Pilots did not sign out Stratus devices when taking them because they were not considered a sensitive item.



Figure 21. Stratus receivers in the B Company flight planning room.

3.4 External Lights

A boxed note near DAA on the DC helicopter route chart stated "HEAVY VFR TRAFFIC BELOW 3000' MSL. ANTICOLIISION, POSITION/NAVIGATION AND/OR LANDING LIGHTS SHOULD BE ON" [ATTACHMENT 9] A battalion standardization pilot stated that when flying the DC helicopter routes at night, aircrews activated the landing light, navigation lights, and flashing anti-collision lights.⁴³ A photo of a 12th Aviation Battalion CEFs-equipped Blackhawk with these lights illuminated is shown in

⁴² Information available at [STRATUS ADS-B RECEIVERS - Stratus By Appareo](#).

⁴³ Interview with Matthew Ingalls, p. 47.

figures 22 through 25. The upper and lower anticollision lights illuminated briefly every two seconds, in alternating fashion.



Figure 22. Front view of a UH-60L with landing and navigation lights illuminated.



Figure 23. Front-left view of UH-60L with landing and navigation lights illuminated.



Figure 24. Left view of UH-60 showing landing light (A), left navigation light (B), flashing lower anticollision beacon (C), and flashing upper anti-collision beacon (D).

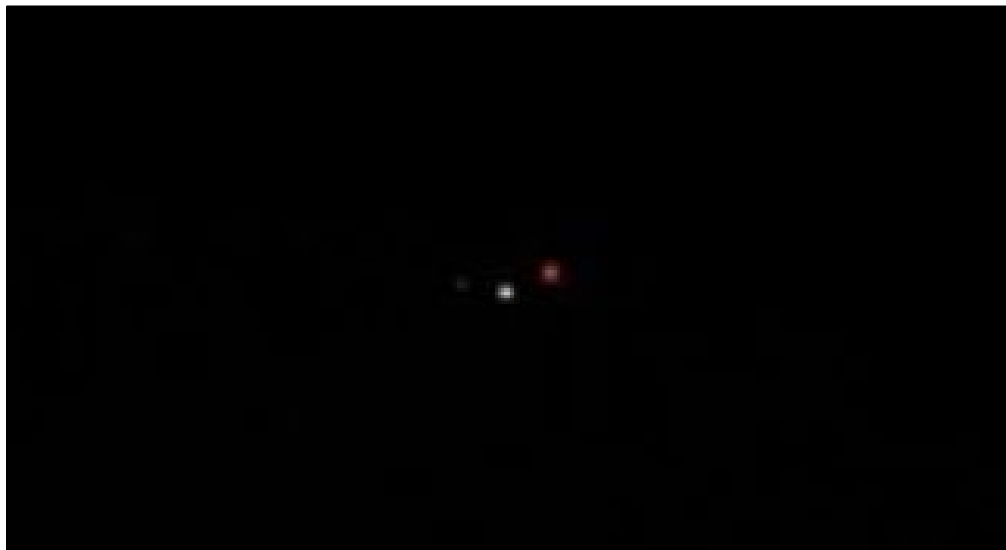


Figure 25. Front left side of the helicopter viewed at night from a vantage point located about 0.5 nm away and 140 feet below the helicopter (flashing anticollision lights are not shown).

3.5 Cockpit Instruments Found in the Wreckage

The helicopter operations and human performance group documented helicopter cockpit displays and controls recovered from the wreckage. This information is contained in Attachment 10, however three examples are displayed in figure 26.

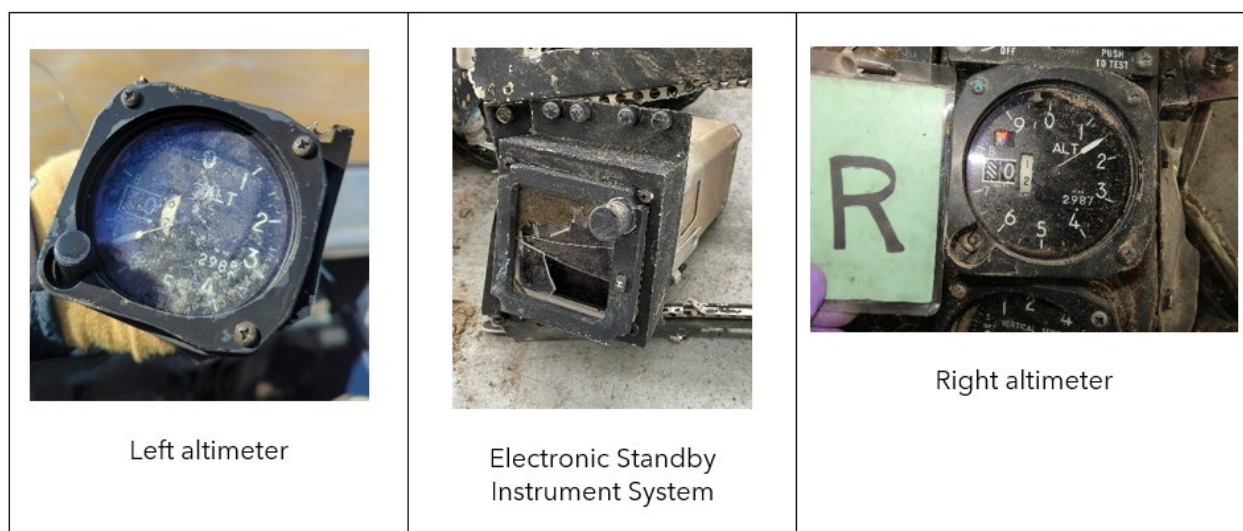


Figure 26. Example displays recovered from the helicopter wreckage.

3.6 Weight & Balance Information

The helicopter operations and human performance group estimated the weight and balance of the accident helicopter at the time of takeoff (table 8) and at the time of the accident (table 9) using the weights listed below, aircraft data, and calculation tools. The group also documented the helicopter's weight and balance limitations (Table 10).

Crew weights:

- IP 220 lbs
- PI 155 lbs
- CE 184 lbs

Personal gear:

- +15 lbs

Planned fuel burn:

- 900 lbs/hr (per the performance planning card for the flight)

Table 8. Estimated weight and balance at takeoff.

Item	Weight (lbs)	Moment/1000 (inches)
Basic Aircraft Weight	12933.8	4612.13
Crew (Pilot & Copilot)	406.0	92.16
Internal fuel (3589 gal)	2405.0	1012.00
External fuel (CEFS)	1795.0	574.40
Crew Chief (STA 262)	199.0	56.11

Laptop & case	8.0	2.10
Survival Kits	20.0	5.64
Tool Box	85.0	35.77
LPU'S	20.0	6.58
Total Corrections	692.0	221.37
Totals	18563.8	6618.26
Helicopter CG		356.51

Table 9. Estimated weight and balance at the time of collision.

Item	Weight (lbs)	Moment/1000 (inches)
Basic Aircraft Weight	12933.8	4612.13
Crew (Pilot & Copilot)	406.0	92.16
Internal fuel (3589 gal)	2405.0	1012.00
External fuel (CEFS)	0.0	0.00
Crew Chief (STA 262)	199.0	56.11
Laptop & case	8.0	2.10
Survival Kits	20.0	5.64
Tool Box	85.0	35.77
LPU'S	20.0	6.58
Total Corrections	692.0	221.37
Totals	16768.8	6043.86
Helicopter CG		360.42

Table 10. Weight and balance limitations.

Limitations	MaxGW (lbs)	Min CG (inches)	Max CG (inches)
	22000	342.1	362.1

4.0 Medical Information

All three of the accident helicopter crew members held current US Army medical certifications. For information about these certifications, crew medical histories, and the results of the crew's post-mortem examinations, see the medical group chairman's factual report.

5.0 Tests and Research

5.1 Night Vision Goggles

Night vision goggles (NVGs) were optoelectronic devices that consisted of an image intensifier tube (usually mounted in pairs), a protective housing, and a mounting system that allowed low ambient light levels to be amplified. They presented an image to the user that more closely resembled vision during the day, allowing operations to be performed in very low lighting (e.g., starlight). NVGs also converted near-infrared light into visible light which can then be seen by the wearer. The Army used various models of NVGs. The accident crew was equipped with ANVIS AN/AVS-6 NVGs (figure 27). A US Army web site provided the following information about ANVIS AN/AVS-6 NVGs:⁴⁴

Aviator's Night Vision Imaging System (ANVIS) is a third-generation, helmet-mounted, direct-view, image-intensification device. It enables Aviators to operate more effectively and safely during lowlight and degraded battlefield conditions.

The low-light sensitivity represents a 35 to 40 percent improvement over the earliest ANVIS. Additionally, the gated power supply enables operation at significantly higher light levels than any of the previous designs. The ANVIS is capable of operating for 22 hours on two (2) AA Alkaline batteries and 34 hours on two Lithium AA batteries. The Low Profile Power Pack (LPPP) encompasses a primary and alternate capability, which allows Aviators to access a safety backup (alternate) of two additional AA batteries.

Specifications:

- Weight (maximum): 1.3 pounds with batteries
- Magnification: 1x
- Operational time: 22 or 34 hours, depending on battery type
- Power: Low-profile battery pack or from aircraft-supplied power

⁴⁴ www.peosoldier.army.mil/Equipment/Equipment-Portfolio/Project-Manager-Soldier-Warrior-Portfolio/Aviators-Night-Vision-Imaging-System/
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Figure 27. Example AN/AVS-6 NVGs. (Source: US Army Training Materials).

The NVGs had an automatic brightness control (ABC) feature. This feature adjusted light amplification to adapt to the ambient conditions. The gain was more sensitive in very low light conditions and less sensitive in brighter conditions. Changes in gain occurred very quickly – on the order of milliseconds. This presented the user with a relatively constant brightness level. The NVGs had a bright source protection (BSP) mode that quickly reduced gain when presented with a very bright light source to protect the NVG from damage.

Pilots' NVGs were attached to a helmet along with a power source (battery pack). Counterweights were attached to the rear of the aviator's helmet to offset the weight of the NVGs. The binocular system could be rotated up on its mount to remove the NVG from the wearer's view. This allowed rapid transition from "aided" to "unaided" vision (figure 28).



Figure 28. A 12th Aviation Battalion pilot wearing AN/AVS-6 helmet-mounted NVGs.

Army training materials indicated that ANVIS AN/AVS-6 NVG tubes provided a 40-degree field of view. NTSB investigators donned 12th Aviation Battalion air crew helmets equipped with NVGs. The image seen through both NVG tubes appeared circular when properly adjusted. Investigators measured the width, height and distance to a wall reference and confirmed that the field of view was approximately 40 degrees (Attachment 11).

Green phosphor ANVIS AN/AVS-6 NVG images provide monochrome green images with approximately 20/40 acuity. White phosphor ANVIS AN/AVS-6 NVGs had improved light amplification capabilities and resolution. They presented a white-colored monochrome image permitting 20/25 acuity (figure 29). According to 12th Aviation Battalion personnel, there was no policy requiring crews to use one type or the other, and there was also no policy requiring crews on the same flight to use the same type (white or green).

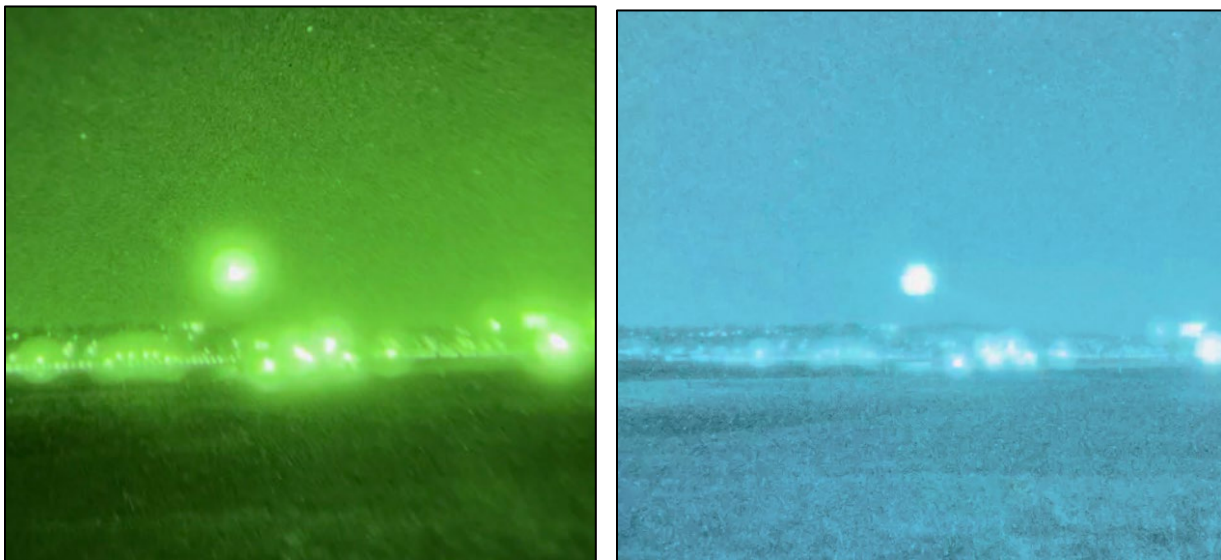


Figure 29. Green (left) and white (right) phosphor ANVIS-9 images of a CRJ on approach to runway 33, viewed from the surface of DCA near the approach end of runway 22.

A standardization pilot from the TAAB stated that it was somewhat challenging to use the NVGs in Washington, DC, where helicopter route 1 met route 4, due to the prevalence of cultural lighting and related dimming of NVG images. However, he did not consider the difficulty “prohibitive”. He said crews would sometimes request advance approval to fly through the area unaided if there was a lot of moonlight.⁴⁵

The helicopter operations and human performance group observed the “evening rush” of commercial traffic at DCA from the top of a building on the Southwest DC Waterfront (elevation 157 feet). This building was located 2.3 nm north of the approach end of runway 1. The observation was made when the airport was in a north landing configuration (airplanes were approaching from the south). The observation was conducted using white phosphor ANVIS 9 NVGs. ANVIS 9 NVGs have the same optical performance as ANVIS AVS/AN-6 NVGs.⁴⁶ An image taken through one of the NVG tubes using an Apple iPhone is shown in figure 30. Multiple airplanes are shown approaching runway 1 at DCA. They are visible above the horizon in the center of the photo. These airplanes appear as point sources of light in the sky without any obvious aircraft structure. These point light sources were surrounded by “halos”, as were many lights on the ground.

Four NVG cases were recovered from the accident area (figure 31). Two were marked with blue tape. 12th Aviation Battalion personnel told investigators that blue tape was indicated NVGs that contained white phosphor. 12th Aviation Battalion confirmed that the NVGs with serial numbers 6307839 and 6301713 contained white phosphor, and the NVGs with serial number 6307927 and 6303678 contained green phosphor. A hand-written receipt (Form 2062) filled out by the IP indicated that the IP checked out all four of these devices before he departed on the accident flight. The first three cases shown in figure 31 (left to right) were empty except for some lens caps. The fourth case contained broken NVG components.

The group also collected three crew helmets (figure 32) and multiple broken NVG components from the wreckage area. Although the group was able to identify which helmets belonged to which crew members, and although some of the NVG components bore serial numbers matching the numbers for the devices that were checked out by the IP and the numbers on the NVG cases, investigators were unable to determine which NVGs had been mounted on which helmets because there were no matching broken wires or fracture surfaces (Attachment 12).

⁴⁵ Interview with CW4 Matthew Ingalls, p. 13.

⁴⁶ Email communication with the Chief, US Army DAAF Detachment Branch, Air Integration & Engineering Division, Engineering & Systems Integration Directorate, DEVCOM C5ISR Center, on March 10, 2025.

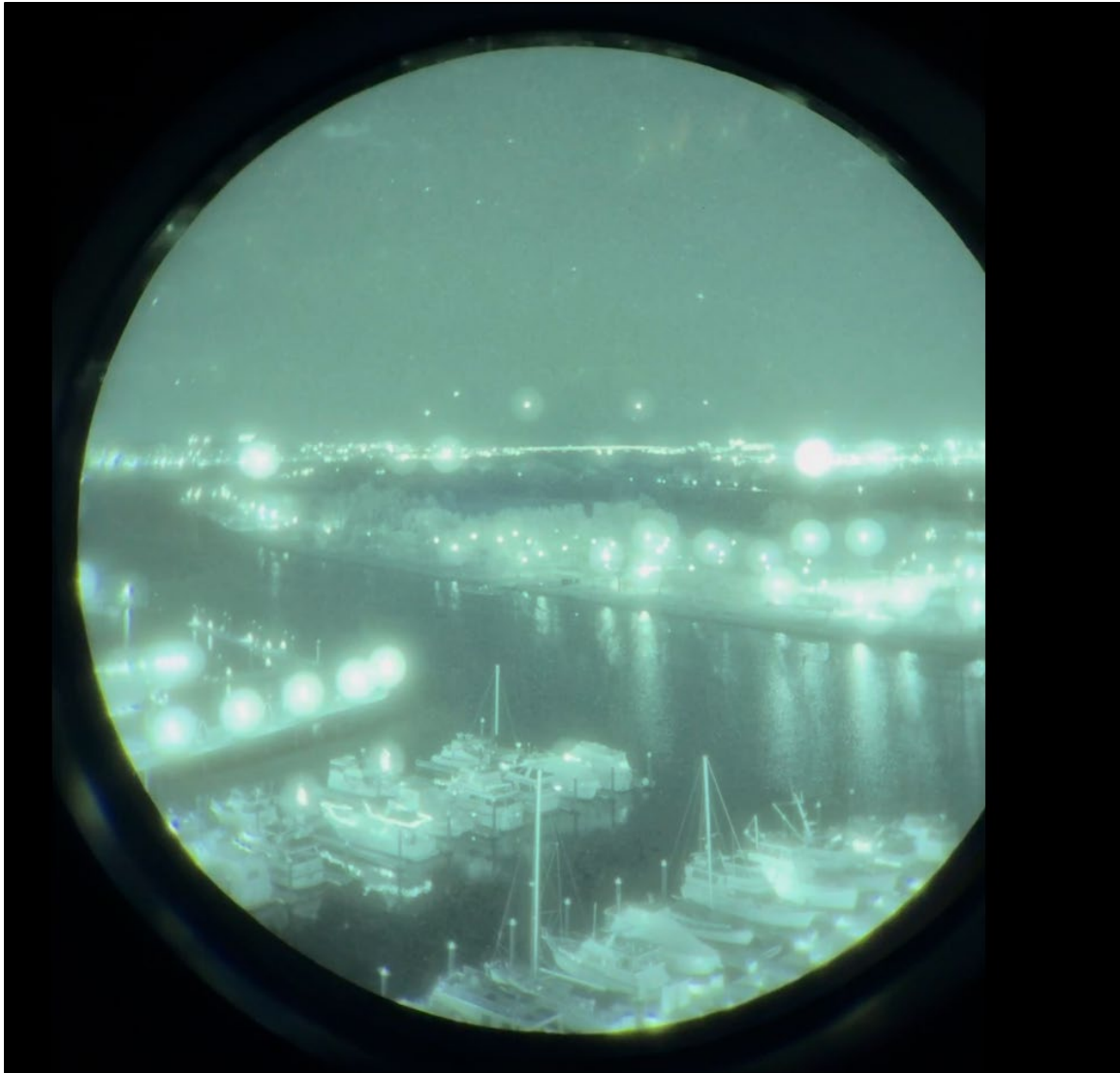


Figure 30. White phosphor NVG view of airplanes approaching DCA runway 1.



Figure 31. Empty NVG cases recovered from the wreckage.



Figure 32. Crew helmets, NVG components, and other electronic components from the wreckage.

5.2 UH-60L Simulator Observations

The helicopter operations and human performance group observed controls, displays, and standard operating procedures in a US Army Black Hawk Aircrew Trainer (BAT) at DAA. According to a US Army web site:

BAT is a highly immersive home-station UH-60L flight training device (FTD), comprised of a state-of-the-art collimated visual system (main display: 200° horiz. x 45° vert., with seven 2-megapixel projectors), a complete UH-60L cockpit, an Instructor Operator Station (IOS), and a vertically expandable container.⁴⁷

The BAT cockpit and aerodynamic model were configured as a UH-60L with the CEFS. A photo of the BAT cockpit is shown in figure 33.

⁴⁷ www.avmc.army.mil/Success-Stories/BAT/, downloaded March 31, 2025.
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Figure 33. Interior view of the Black Hawk Aircrew Trainer.

5.2.1 NVG Adjustment for Viewing Instrument Panel

According to Army personnel, pilots adjusted the NVGs before flight to ensure proper focus and comfort and to provide enough space to see “under” them to view aircraft instruments. A helicopter operations and human performance group member who was 6’4” tall sat at the left pilot station with the seat full aft. The group member’s height was lowered until their eyes were positioned 38” (965 mm) above the simulator cab floor, which was equivalent to the eye height of a 12th AB pilot in the right seat who adjusted his seat to a “normal” height for flying. The group member donned NVGs and lowered them for use. The group member then adjusted the tubes forward until the instrument panel was fully visible below the tubes. A caliper was used to measure the distance between the NVG eye piece and the outer corner of the group member’s eyelids. The resulting measurement was 54mm (figure 35).



Figure 34. Forward adjustment that permitted full view of the instrument panel.

The group member was able shift their gaze from looking at the external scene through the NVGs to looking at the flight instruments below the NVGs without head movement. The portion of instrument panel that was visible below the goggles is shown in figure 36. A photo was taken from the group member's goggle position, and the group member used computer software to sketch the approximate field of view through the NVGs when looking straight ahead through them (figure 37).



Figure 35. View of the instrument panel when looking below the NVG's with the seat full aft and the eyepiece 54mm in front of the wearer's eye.

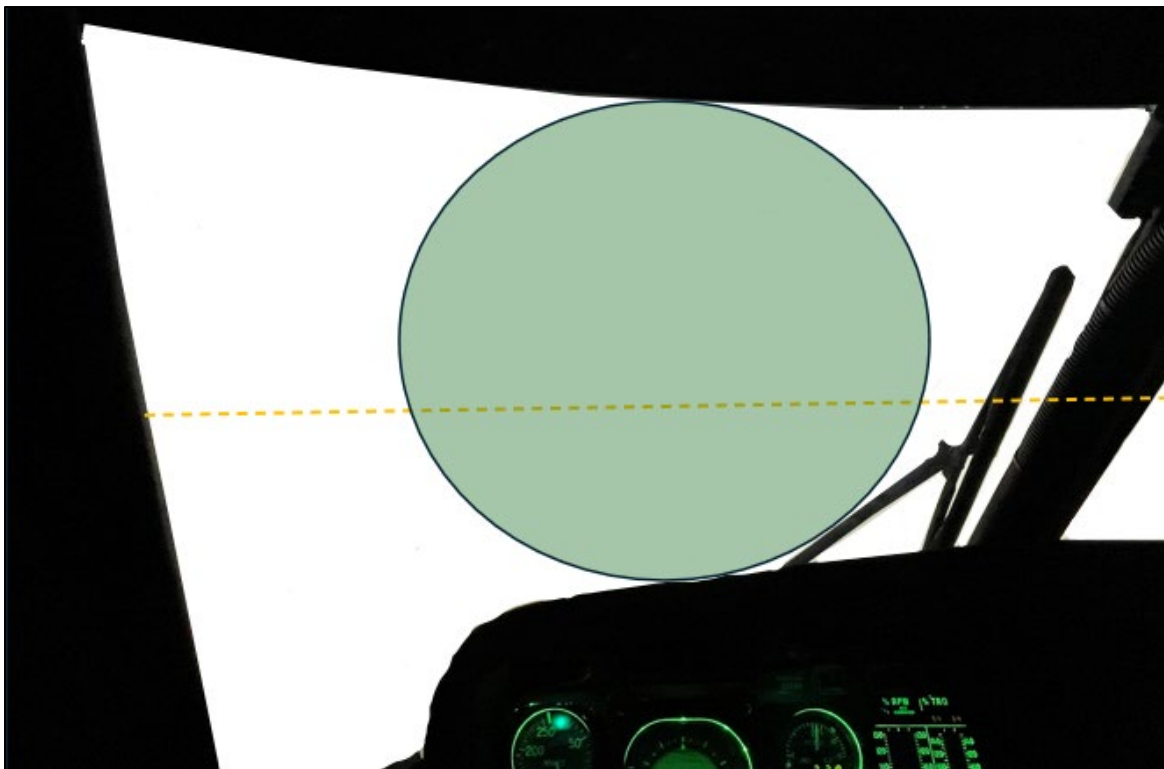


Figure 36. Sketch of the left pilot's NVG field of view when looking straight ahead.

5.2.2 Cross-Cockpit Visibility of Altitude Instruments

With goggles removed, group members observed the visibility of the altitude instruments on the instrument clusters in front of the adjacent pilot. The locations of the barometric and radar altimeters are shown in figure 38.

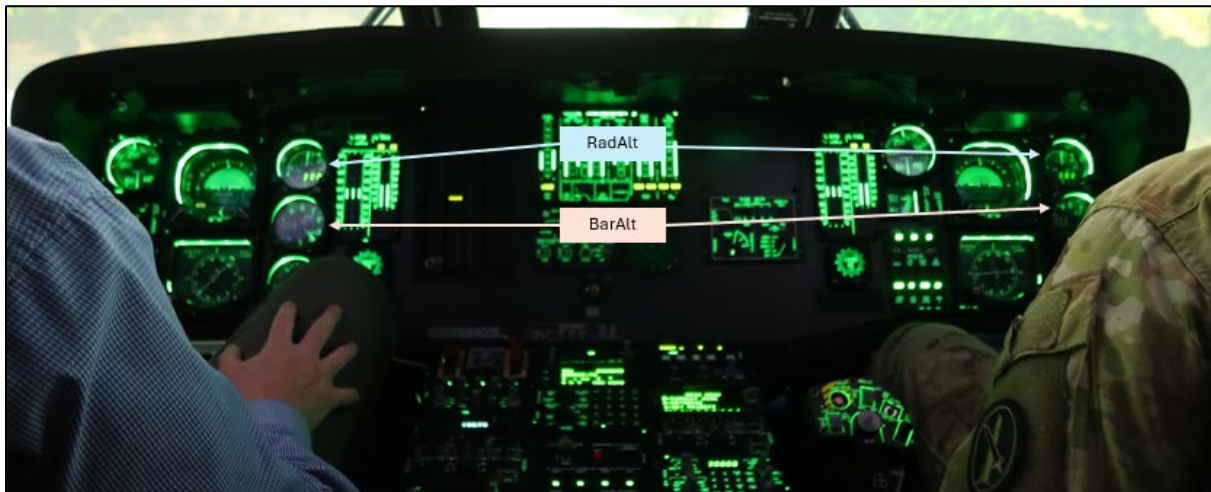


Figure 37. Location of barometric and radar altimeters.



Figure 38. Left-seat pilot's view of right seat altimeters.

A view of the right seat altimeters from the point of view of the left seat pilot is shown in figure 39. The left seat pilot could not easily observe altitude indications on the opposite side of the cockpit. Head rotation and leaning were required. The left seat altimeters were closer to the center of the cockpit, which made it somewhat easier for the right seat pilot to observe the left-seat altimeter indications.

5.2.3 Radar Altimeter

The UH-60L radar altimeter was equipped with adjustable upper and lower altitude bugs. Investigators observed in a UH-60L simulator that when the radar altitude dropped below the lower bug, an amber "LO" indication illuminated on the face of the radar altimeter, and a green "DH" indication illuminated above the vertical situation indicator (VSI), as shown in figures 40 and 41.



Figure 39. "LO" and "DH" indications (yellow arrows) that appeared when radio altitude decreased below the radar altimeter's lower altitude bug.



Figure 40. Location of the "LO" and "DH" indications on the left instrument panel.

A TAAB standardization pilot said that he did not use radar altimeter bugs when flying the DC helicopter routes, and that no standard operating procedure required pilots to use the radar altimeter in that context.⁴⁸

5.2.4 Subjective Workload Estimation During Low Level Flight

A qualified and current TAAB UH-60 pilot sat at the right pilot station and operated a simulated flight between the Arlington Memorial Bridge and the accident location, following the transition from route 1 to route 4 at 200 feet msl.⁴⁹ The pilot wore NVGs and the environmental lighting conditions were set to nighttime. Simulated wind approximated the accident conditions: 300 degrees at 14 knots gusting to 23 knots. The simulated flight was 2 minutes 55 seconds in duration.

After completing this flight, the pilot provided a subjective workload rating using the Bedford Workload Rating Scale (which ranges from 1 "workload insignificant", to 10 "tasks abandoned, pilot unable to apply sufficient effort"). The pilot estimated that workload during this simulated flight as a 4, "insufficient spare capacity for easy attention to additional tasks." He noted that the aerodynamic effects of CEFS and the programmed wind conditions made it more challenging than usual

⁴⁸ Interview with CW4 Matthew Ingalls, p. 49.

⁴⁹ The pilot was an experienced Army aviator with 646 total hours, 534 in the UH-60M, 38 in the UH-60L, 74 in the UH-72, and 189 hours of NVG time (UH-60L, UH-60M, and UH-72).

to maintain altitude, airspeed, and trim. These factors required him to frequently crosscheck the flight instruments and make corrections. He said that if there had been no wind he likely would have considered the workload a 2, "workload low", and if there had been no wind and the flight had been in the daytime, he would have considered it a 1.

The pilot stated that during a single glance down he might look at his attitude indicator, radar altimeter (because he was close to the ground), vertical speed indicator, or airspeed. He stated that a typical pattern was airspeed, attitude and external view, followed by attitude, radar altimeter and external view, followed by VSI, barometric altimeter and external view. A video of the pilot's left eye was recorded during the simulated flight and retained for subsequent review. This review indicated that the pilot looked below the goggles 19 times during the flight, an average of once every 9.2 seconds.

5.2.5 ForeFlight Traffic Display

A helicopter chart of the DCA terminal area was displayed on one of the 12th Aviation Battalion iPad Minis using ForeFlight. Live traffic information was turned on. A screen shot was taken showing a CRJ approaching runway 33. This screenshot was saved on the iPad as a still photo (figure 44). The iPad was placed on the right seat pilot's left thigh as he was flying from Hains Point to the accident location on helicopter route 4 at 200 feet msl. The pilot was instructed not to look at the iPad until told to do so. A group member in the left seat video recorded the right-seat pilot's left eye to determine when the pilot was looking through the NVGs versus looking down at the iPad. As the helicopter approached the accident location, the group member issued a warning about a potential conflicting CRJ ahead and asked the pilot to locate the target on the iPad and announce when he had ascertained its position. The group member's warning and instruction lasted 7 seconds in duration. The pilot verbalized that he had identified the target 3 seconds later and the simulator was paused.

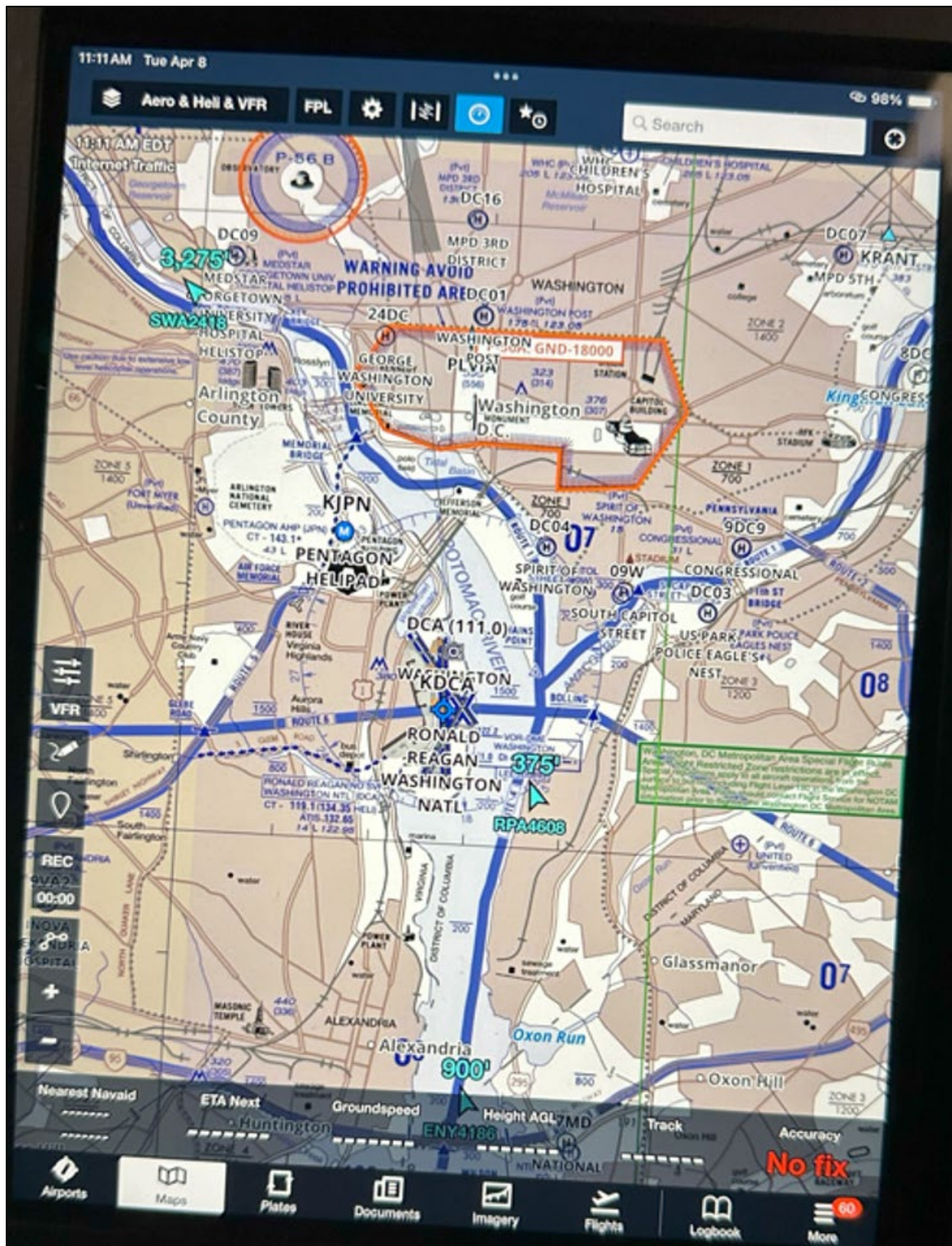


Figure 41. Screen shot of the DC helicopter route chart showing traffic approaching runway 33.

The group's human performance investigator reviewed the resulting video with the pilot immediately after the simulator was paused and asked the pilot when he first acquired the target and understood whether it was a conflict. The pilot reported that he had done this during his second glance at the iPad, which video indicated was simultaneous with his verbalization that he had completed the task.

Video revealed that the pilot tilted his head down each time he viewed the iPad (figure 43). The group's human performance investigator sat in a chair with the same iPad on his thigh while looking straight ahead and confirmed that it was necessary to tilt one's head down to look at the iPad because it was outside the range of eye movement when looking straight ahead.



Figure 42. Pilot's head tilt when acquiring traffic on the iPad.

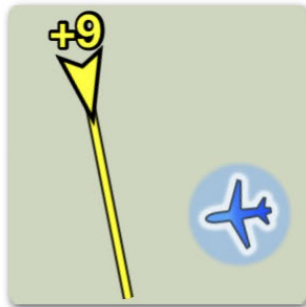
The pilot told investigators he would have been able to locate the traffic more quickly if it had been depicted in red (as would happen during an actual ForeFlight traffic warning) and if his ownship icon had also been shown on the display. An excerpt of the 2024 ForeFlight Mobile Pilot's Guide showing traffic warning symbology is reproduced in Figure 44.⁵⁰

⁵⁰ ForeFlight Mobile Pilot's Guide. (2024), p. 830.
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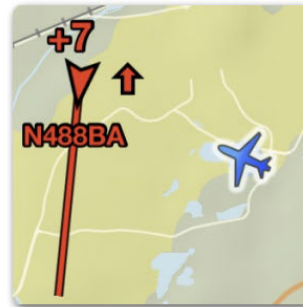
Traffic Cautions are issued for targets that are less than 45 seconds from being within 2.0 NM and +/- 1,200' of ownship position. When a caution is issued, the target icon is highlighted **YELLOW**.

Traffic Warnings are issued for targets that are less than 25 seconds from being within 1.3 NM and +/- 1,200' of ownship position. When a warning is issued, the target icon is highlighted **RED**, and a **Traffic Alert** is activated if the setting is enabled.

Traffic targets remain highlighted for 15 seconds after they no longer exceed the thresholds.



Traffic Caution
within 2.0 NM and
+/- 1,200' or will be
within 45 seconds



Traffic Warning
within 1.3 NM and
+/- 1,200' or will be
within 25 seconds

Figure 43. ForeFlight Mobile Pilot's Guide illustration showing a traffic conflict.

6.0 Aircrew Duties and Training Requirements

6.1 Aircrew Duties

US Army aircrew duties are defined as follows in US Army Regulation AR 95-1.

Section II

Flight Crewmembers

4-18. Flight crews

Unit commanders must establish, in writing, formal flight crew qualification and selection programs. Programs will contain qualification and selection criteria and evaluation requirements. IP/operators and safety officers will aid commanders in the selection process. Rated and nonrated personnel will be designated, in writing, by the commander, specifying the duties and flight crew stations that they are authorized to occupy per TC 3-04.11.

4-19. Pilot-in-command

The PC will be—

- a.* The individual responsible and having final authority for operating, servicing, and securing the piloted aircraft.
- b.* Selected per paragraph 4-18 for each flight or series of flights.
- c.* Qualified, current, and designated RL1 in the aircraft mission, type, design, and series.
- d.* Listed in the flight plan or unit operations log.
- e.* Responsible for crew and passenger briefings.
- f.* At a crew station with access to the flight controls.
- g.* Approved per the mission approval process before each mission. UTs, IPs, SPs, MEs, or IEs, when performing duties from other than the pilot or CP station will participate in the mission approval process.

4-20. Air mission commander

When two or more aircraft are operating as one flight, the unit commander will designate one of the rated aviators of the flight as an air mission commander to be in command of all aircraft in the flight. The designation of air mission commander is an assignment of command responsibility and is not an aircrew duty assignment. Air mission commanders will be chosen based on recent aviation experience, maturity, judgment, their abilities for mission situational awareness, the understanding of the commander's intent, and not necessarily upon rank or grade. Air mission commanders will participate in the mission approval process and may receive the final mission approval for all crews in the flight.

4-22. Co-pilot

The CP will assist in the performance of tasks as directed by the PC and is an aviator who—

- a.* Is at a crew station with access to the flight controls but is not qualified or current in the aircraft being flown.
- b.* Is at a crew station without access to the flight controls and performing duties required for the mission.
- c.* Is performing CP duties at other than a flight crew station and is undergoing training or evaluation conducted by an IP, SP, IE, UT, or ME.

4-23. Unit trainer

The unit commander may appoint UTs to conduct specialized training to assist in unit training programs. UTs are prohibited from conducting emergency procedures training in the aircraft/UAS. UTs are forbidden to evaluate aircrew training tasks. Commanders may authorize rated aviators and UAS operator UTs to conduct duties from any crew station. They may also authorize UTs to validate successful completion of required training; for example, border and corridor qualifications, local area orientation, and other locally directed requirements. When performing UT duties, the UT must be qualified and current in the aircraft/UAS being flown.

4-24. Instructor pilot

The IP will train and evaluate all personnel in designated aircraft per approved aircrew training tasks and TC 3-04.11. To become qualified as an IP for helicopters or airplanes, an aviator must be qualified as a PC and must complete one of the following:

a. Helicopters.

(1) A course of instruction for IPs at an authorized Aviation Proponent School in the mission, type, and design aircraft in which IP duties are to be performed.

(2) An IP equivalency evaluation administered by an SP selected by USAACE, DES (ATZQ-ES) in the mission, type, and design aircraft in which IP duties are to be performed. Commanders will coordinate with DES before submitting a request for equivalency evaluation through the chain of command to DCS, G-3/5/7 (DAMO-AV) for approval.

(3) In the absence of a course of instruction for IPs at an authorized Aviation Proponent School for the aircraft, ACOM, ASCC, DRU commanders, or the DARNG may approve an additional IP qualification to be conducted locally for helicopter IPs who are qualified per paragraph 4-26a or 4-26b of this regulation.

4-26. Standardization instructor pilot

The SP may train and evaluate all personnel in the designated aircraft per approved aircrew training tasks. SPs have technical supervision of the unit aviation standardization program as specified by the unit commander. They advise the commander at all levels of aviation standardization within the command.

a. Qualified IPs per paragraph 4-24 will be designated in writing as SPs by unit commanders and be qualified and current in at least one of the aircraft assigned to the unit. Commanders may authorize SPs to instruct and evaluate from any station.

b. An SP will administer training and evaluation in specified SP duties as described in the ATM and 3-04.11.

c. For additional aircraft, the aviator must meet the following requirements:

(1) Designated as SP in primary aircraft.

(2) Qualified in accordance with paragraph 4-24 for the additional aircraft.

(3) Successfully complete an SP evaluation.

4-31. Crew chief (Manned)

The CE is a crewmember that must perform duties aboard an aircraft that are essential to its operation and/or specific flight mission. They will be—

a. In an assigned flight position by MTOE and/or TDA per AR 600-106 or as required by the contract.

b. Selected per paragraph 4-18 for each flight and/or series of flights.

c. MOS-qualified.

d. Trained to perform crew duties per TC 3-04.11 and approved aircrew training tasks.

e. Listed on the flight briefing and flight plan or unit operations log.

4-34. Crewmember flight instructor

The FI trains and evaluates other crewmembers and noncrewmembers in their designated aircraft system or aircraft mission per the MTL or nonstandard aircraft ATM as applicable. To become qualified as an FI, the crewmember must be qualified in accordance with this regulation and complete one of the following—

a. Complete a course of instruction for FIs at an authorized Aviation Proponent School.

b. Complete an FI equivalency evaluation administered by an SI selected by USAACE, DES (ATZQ-ES) in the type aircraft in which the FI duties are to be performed. Commanders will coordinate with DES for an equivalency evaluation. An equivalency evaluation only applies to MOSs with an authorized Aviation Proponent School.

c. If an authorized Aviation Proponent School does not exist or is not available for a specific aircraft or MOS, commanders may select a highly experienced crewmember to perform FI duties. The selected individual will be trained and evaluated by an IP, SP, or an SI per the approved aircrew training tasks and TC 3-04.11.

d. Crewmembers with an authorized N1 ASI who change MOS will retain the N1 identifier and are not required to complete an Enlisted Flight Instructor Course for their new aircraft MOS.

4-35. Crewmember standardization instructor

The SI is a crewmember that trains and evaluates FIs, and SIs. They also assist the unit SP with supervision and maintenance of the unit ATP.

a. To be designated by the commander as an SI, the FI must complete a course of instruction for FIs at an authorized Aviation Proponent School.

b. To be designated by the commander as an SI, the FI must complete an SI evaluation administered by an SI/IP/SP in the type aircraft in which the SI duties are to be performed.

c. Battalion/Brigade/Task Force SIs are authorized to perform SI duties on all aircraft assigned to their unit for which they have completed an aircraft qualification.

d. If an authorized Aviation Proponent School does not exist or is not available for a specific aircraft or MOS, commanders may select a highly experienced crewmember to perform SI duties. The selected individual will be trained and evaluated by an IP/SP/SI per the approved aircrew training tasks and TC 3-04.11.

e. Crewmembers with an authorized N1 ASI who change MOS will retain the N1 identifier and are not required to complete an Enlisted Flight Instructor Course for their new aircraft MOS.

A battalion standardization pilot said that if he was conducting an NVG APART flight with another pilot in the local area he would rely on a CE for airspace surveillance, identification of hazards, and fuel consumption checks.⁵¹ He said that the window in front of the CE was normally left open during a low altitude VFR flight, and that if a CE leaned out the window they had “full left, right, up and down visibility.”⁵² CEs used standard terminology to call out traffic: clock direction for azimuth, and distance in nautical miles. He said it was “very common” for a CE to sit on the same side as the pilot who was being evaluated to monitor their instruments and “help with that workload”.⁵³

6.2 12th Aviation Battalion Flight Scheduling

The battalion generated a weekly flight schedule that incorporated all assigned flying and simulator activities for Alpha, Bravo, Charlie, and National Guard companies. Each day consisted of roughly 20 to 30 scheduled events with crews assigned. About half the scheduled crews were assigned an active mission line. The remaining half were usually participating in training flights. Most training flights were allocated a 3 to 4-hour block of flight time. The flight schedule was signed by the battalion commander.

6.3 Crew Rest Policy

A battalion standardization pilot stated that pilots normally received at least 10 hours of rest between duty periods. If they received less than 10 hours the risk level increased, and the mission required a signoff from the battalion commander. If they received less than 8 hours, the mission was considered high risk and required higher level signoff.

A B Company SFC was asked about CE duty period limits, and he said, “we try to limit it to 10 hours of duty day, and we’ll go out to about 12.” He added that time beyond 12 hours required approval for an extension and they would not work “more than an hour, at most, past that.”⁵⁴

The TAAB SOP. Section 4-8 through 4-14, Fighter Management, described the policies for duty times and rest periods. The following summary applied to non-contingency duty, which was the 12th Aviation Battalion’s status at the time of the mishap. A duty period was not to exceed 12 hours; however, a company commander could extend that period by 2 hours on a case-by-case basis. Rest periods were the 10 hours after completion of the duty period and the start of the next duty period. A

⁵¹ Interview with CW4 Matthew Ingalls, p. 30.

⁵² The standardization pilot stated, however that CEFS would partially restrict the CE’s visibility.

⁵³ Interview with CW4 Matthew Ingalls, p. 31.

⁵⁴ Interview with SFC Wayn Bulson, p. 14.

24-hour rest period, where no duty related activities are performed, should be scheduled every 14 days. The battalion commander could adjust rest periods to 8 hours on a case-by-case basis. Flight hour limitations were 8 hours maximum, 7 hours if any combination of day, night, hood, weather, or NVG, with a maximum of 6 hours of NVG time per 24-hour period.

6.4 Training and Currency Requirements

6.4.1 Army-Specific Training Requirements

6.4.1.1 Flight Activity Category (FAC)

Training Circular (TC) 3-04.11, Commander's Aviation Training and Standardization Program, is used in conjunction with AR95-1 for commanders to develop, manage, and administer a comprehensive aviation training program.

TC 3-04.11 established Flight Activity Categories (FAC) based on task requirements and proficiency required in a particular aircraft to execute a particular mission.

FAC-1 duty positions required a high degree of proficiency and the tactical employment of the assigned aircraft executing mission essential task list (METL)

FAC 2 duty positions required the same level of proficiency in individual and crew task as FAC 1 duty positions, less in company and battalion collective mission tasks.

FAC 3 duty positions were only applied in table of distribution and allowances (TDA) units and modified table of organization and equipment (MTOE) unit position at brigade level and above

FAC 4 designations were only applied to operational flying position above and outside a combat aviation brigade where a brigade level commander or higher had determined there were limited and or no aircraft or simulators available to support FAC 1, 2, or 3.

All flying positions at the 12th Aviation Battalion were FAC 2 positions, as stipulated in the TAAB SOP chapter 2, section 2-3. The TC 3-04.11 directed, and it was noted in the TAAB SOP, that all rated crew members (RCM) with less than 3 years in their initial operational assignment would be assigned FAC 1 flying hour minimums.

The H-60 Series Aircrew Training Manual (ATM) established the minimum flight hour requirements for Army aircraft. For UH-60L model helicopter FAC 1 minimums were 48 hours semiannually, and FAC 2 minimums were 30 hours semiannually.

6.4.1.2 Flight Hour Requirements

The Commander's Task List established the minimum required flight hours for each crew member. Flight hour requirements could be adjusted and/or prorated by the commander to account for such things as a lack of access to a suitable simulator or crew member being in a temporary medical suspension. The FAC 1 flight hour minimum for the UH-60L was 96 hours annually, and the FAC 2 flight hour minimum was 60 hours annually.

Army aircrew currency requirements were specified in Army regulation AR95-1. An aviator had to maintain currency in both the aircraft and under the NVG mode of flight. To maintain aircraft currency, an aviator had to log at least 0.1 hours every 60 days. To maintain NVG mode of flight currency, an aviator had to log a full 1.0 hours or more every 60 days. Crewmembers who did not meet currency requirements were required to complete a PFE. An excerpt of AR95-1 is presented below.

4-16. Currency

Currency requirements in any one series aircraft will satisfy the currency requirement for all aircraft with similar operating and handling characteristics (series group). Separate currency is required for all other aircraft.

a. (Manned) To be considered current in the aircraft, individuals must take part in flight once every 60 days at a crew station with access to the flight controls. Individuals must perform designated duties in a crew station authorized on the Commander's Task List. If 60 days have elapsed since the last flight, the crewmember will be administered a PFE in the aircraft. NVD currency will be per TC 3-04.11 and the appropriate MTL.

b. (Unmanned) To be considered current, an operator must perform a flight at the AO station once every 60 days. The flight may be conducted with the aircraft or in a simulator. If 120 days have elapsed since the last aircraft flight, the operator will be administered a PFE in the aircraft per the appropriate MTL. Simulators may not be used to reestablish currency.

Aircrew currency requirements were also affected by unit requirements. 12th Aviation Battalion required pilots to have a minimum of 1 hour night unaided every six months, and 3 hours of simulated IMC using a view limiting device or actual IMC every six months.

6.4.1.3 IP Flight Hour Requirements

The IP's Commander's Task List for the time period between 01 December 2024 and 30 November 2025, required him to have 60 hours of flight time, 2 hours of night unaided time, 18 hours of NVG time, 6 hours instrument time (hood/weather), and 12 hours of simulator time annually.

For the period beginning Oct 16, 2023, through November 14, 2024, the IP accumulated 276.0 flight hours, exceeding the FAC 1 annual flight hour requirement. The IP was assigned to B Company, at the 12th Aviation Battalion on October 2, 2024, and had accumulated an additional 25.5 hours as of January 28, 2025.

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Flying-hour requirements for the IP:

Total Aircraft Hours	30 hours / 6 months
Total Simulator Hours	12.0 Annually
Night Unaided:	1.0 hours / 6 months
NVG Hours:	9.0 / 6 months
Hood/Weather Hours:	3.0 / 6 months

6.4.1.4 PI Flight Hour Requirements

The PI's Commander's Task List for the period March 1, 2024, to February 28, 2025, required 38 hours of flight time, 1.2 hours of night unaided time, 10.5 hours of NVG time, 3.5 hours instrument time (hood/weather), and 8.5 simulator hours. The PI's flight time requirements had been prorated by the company commander to account for 5 months of temporary medical suspension during the period.

Since the 12th Aviation Battalion was the PI's initial operational assignment, she was required to maintain FAC 1 flight hour minimums for the first three years⁵⁵. She was assigned to B Company, 12th Aviation Battalion on July 19, 2021. Her 3-year anniversary date was July 19, 2024, and she was assigned FAC 2 flight hour minimums when her second semi-annual period began on September 1, 2024.

She was established RL 1 January 06, 2022. She had one month of flight hour minimums required for the month of February 2022. Due to a reduction in RL status in February, she was unable to complete her flight hour requirements. She was granted an extension to complete flight minimums of 0.5 hours of instrument (hood/weather) time and 0.8 hours of simulator time. Extension requirements were complete on March 16, 2022 and she was redesignated RL 1.

For the period from March 1, 2022, through February 28, 2023, the PI accumulated 129.7 flight hours and 20 simulator hours, exceeding the FAC 1 requirement of 96 hours.

For the period from March 1, 2023, through February 28, 2024, the PI accumulated 87.5 flight hours and 14 simulator hours. Flight hour and simulator requirements were prorated for the loss of two months due to temporary medical suspension. Requirements were met during this period.

For the period from March 1, 2024, through February 28, 2025, the PI accumulated 34.7 flight hours and 10 simulator hours. The first semi-annual period

⁵⁵ TC 3-04.11 Sec 7-25, and TAAB SOP Sec 2-3.

was prorated 5 months due to temporary medical suspension. Requirements were met for this first semi-annual period. Not including the accident flight hours, the PI had 21.5 hours towards her now FAC 2 minimums of 30 hours semi-annually.

Flying-hour requirements for the PI:

Aircraft total hours:	30 hours / 6 months
Total Simulator Hours:	8.5 Annually
Night Unaided:	1.0 / 6 months
NVG Hours:	9.0 / 6 months
Hood/Weather Hours:	3.0 / 6 months

Asked whether pilots in the 12th Aviation Battalion had experienced difficulty getting enough flight time to remain current, a brigade standardization pilot said it was very rare for a pilot to go out of currency for their assigned aircraft.

6.4.1.5 CE Flight Hour Requirements⁵⁶

Flying-hour requirements for the CE:

Total Aircraft Hours:	24.0 / 6 months
Night Unaided:	1.0 / 6 months
NVG Hours:	5.0 / 6 months

6.4.1.6 Readiness Level (RL) Progression

The following information about training requirements for a new pilot arriving at a company is summarized below from US Army Training Circular TC 3-04.11, Commander's Aviation Training and Standardization Program.

When a new pilot arrived at the unit from flight school, they had 14 days to give their flight records to the aircrew training program (ATP) manager (usually the company commander or the standardization pilot in the flight company). Once assigned, a commander's evaluation had to be conducted within 45 days. A commander's evaluation could be just a record review if they were to be designated Readiness Level 3 (RL3). If they were to be designated RL 2, they had to have a records review and an evaluation flight before being so designated. It was uncommon for a new aviator from flight school to be designated anything other than RL 3, especially if they needed to complete an aircraft series transition to the UH-60L.

An aviator was allowed 90 days to complete each phase of RL progression. During RL3 progression training, an aviator worked on basic tasks first, then mission

⁵⁶ UH-60L ATM
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level tasks during the RL2 phase. Once both phases of RL progression were completed (which could take a total of 180 days) a new aviator would be designated RL1. After being designated RL1, the aviator was considered fully mission capable and could fly with any PC. Although a total of 180 days was allowed, it did not always take the full amount of allotted time to achieve the RL1 designation, because the training was performance based.

Once an aviator was designated RL1, they had minimum flight hour and recurring training and evaluation requirements. A new RL1 aviator within three years of graduating flight school, was required to fly 48 hours every semi-annual period. Of those 48 hours: nine had to be flown using NVGs, three had to be flown IMC or simulated IMC (hood/weather), and one had to be flown in night unaided conditions. Other ATP requirements included a written examination, a block of instruction for aircrew coordination training, an annual instrument flight evaluation, and an annual standardization flight evaluation. The standardization flight evaluation was usually conducted under NVG mode of flight to satisfy both the standardization flight and annual NVG flight evaluation requirements on a single flight.

6.4.1.7 Annual Proficiency and Readiness Test (APART)

Rated crewmembers (RCM) were required to complete an annual proficiency and readiness test (APART) consisting of the following parts: a standardization flight evaluation, NVG flight evaluation, annual written examination, and an instrument flight evaluation. The standardization flight evaluation was to be conducted within the framework of the unit's tactical mission. An NVG evaluation was required for the aircraft category in which the crewmember performed duties. The instrument evaluation consisted of all the instrument tasks designated in the master task list (MTL) and they had to be conducted in the aircraft category in which the crew member performed duties.⁵⁷ Task standards described the minimum level of performance required for task proficiency used during readiness level progression, training, and evaluations. The following example task standards were listed in the UH-60L Aircrew Training Manual.⁵⁸

All tasks:

- Do not exceed aircraft limitations.
- Perform crew coordination actions IAW aircrew coordination section in this ATM.
- Crewmembers will comply with all evaluation considerations, warnings, cautions, and notes in the task.

⁵⁷ The Minimum Required Task List (MTL) is a consolidated listing of all tasks that apply to the operation of the airframe covered in the Aircrew Training Manual

⁵⁸ Extracted tasks from the UH-60L ATP

Hover:

- Maintain heading ± 10 degrees (± 20 degrees for automatic flight control system (AFCS) off).
- Maintain altitude, ± 3 feet (± 5 feet for out of ground effect (OGE)). *
- Do not allow drift to exceed 3 feet (10 feet for OGE hover). *
- Maintain ground track within 3 feet.
- Maintain a constant rate of movement appropriate for existing conditions.

Note. These standards (*) required that the other crewmembers announce drift and altitude changes before exceeding the standard.

In-flight:⁵⁹

- Maintain heading ± 10 degrees.
- Maintain altitude ± 100 feet.
- Maintain airspeed ± 10 KIAS.
- Maintain ground track with minimum drift.
- Maintain rate of climb or descent ± 200 FPM.
- Maintain the aircraft in trim $\pm \frac{1}{2}$ ball width

The ATP⁶⁰ allowed for evaluations to be conducted in the “most demanding mode” which most closely replicated the unit mission. In the case of the 12th Aviation Battalion’s mission the most demanding flight mode would be night NVG operations. By combining the PI’s standardization check with the NVG check, it closely replicated the actual mission environment and eliminated the need to complete an additional evaluation flight. Additionally, crewmembers who were designated to fly from both left and right seats would be evaluated in each seat during an APART evaluation.

6.4.1.8 Additional Training Requirements

With respect to local area familiarization, A TAAB standardization pilot stated that pilots who were new to the 12th Aviation Battalion had to complete an online safety course for operating in the Special Flight Rules Area and the Flight Restricted Zone around the capital, and they had to receive a local area orientation flight with an instructor that included operations along helicopter routes 1, 3, 4, 5 and 7. These familiarizations had to be accomplished at day and at night. The night flights had to be performed both aided and unaided. These orientation flights had to be completed before a pilot was signed off to fly in the local area.⁶¹

⁵⁹ In comparison to the FAA Commercial Practical Test Standards for Rotorcraft (FAA-S-8081-16). In-flight: Maintain the appropriate altitude, ± 100 feet (30 meters) and established heading, $\pm 10^\circ$.

⁶⁰ TC 3-04.11, Section 9-2 and 9-3.

⁶¹ Interview with CW4 Matthew Ingalls, p.11.

6.4.1.9 Ground / Academic training

Aircrew coordination training (ACT) was integrated into the Army's training as required by the Commander's Aviation Training and Standardization Program (TC 3-04.11). It had two essential components, an initial qualification followed by annual sustainment training. The initial qualification was conducted during initial flight training. The annual training was conducted at the unit level by qualified standardization personnel and the training materials were updated annually. Standardized guidance for aircrew coordination in the UH-60L was contained in chapter 7 of the UH-60L Aircrew Training Manual and included topics on crew climate, pre-mission planning, situational awareness, decisions and actions communications, crewmembers mutual cross monitoring, advocacy and assertion, and standardized crew terminology. The IP was a qualified air crew coordination instructor for B Company. The most recent ACT was completed by the IP on 12 November, 2024, the PI on 26 January 2024, and the CE on 9 July, 2024.

The battalion conducted monthly aviator academics where various topics and presentations were discussed⁶². Guidance for this training was set forth in the TAAB SOP Section 3-7. Example topics for aviator academics were; pre-mishap plan / DART operations, terrain flight planning and navigation, settling with power/vortex ring state, mission approval and risk management process, air crew coordination training and aeromedical topics.

The battalion maintained an aircrew information reading file (AIRF) system that contained reference material pertinent to aviation operations, standardization and safety.⁶³ The reading file was divided into three categories, new information, quarterly information, and permanent information. The air crewmember was required to check their reading file regularly and read the most recent additions to the file before meeting with a mission briefing officer (MBO) and being issued keys to a helicopter. The reading file was kept on a networked computer site and crewmembers were marked green when their reading file was up to date.

6.4.2 FAA Training Requirements

All pilots operating in the Washington, DC special flight rules area (SFRA) were required to complete a training module describing the rules and restrictions that applied to operations in that area⁶⁴. The IP completed this training on June 12, 2019, and the PI completed this training on August 23, 2021.

⁶² Interview with CW3 Barczak, Battalion standardization pilot.

⁶³ TC 2-04.11, Section 8-93 thru 8-95.

⁶⁴ Activities, Courses, Seminars & Webinars - ALC_Content - FAA - FAASafetyTeam - FAASafety.gov
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7.0 Organizational & Management Information

7.1 Organizational Structure

The accident helicopter and crew belonged to B Company, 12th Aviation Battalion, based at DAA (part of Fort Belvoir, Virginia). The IP and CE reported to the B Company commander, a captain. The captain reported to the 12th Aviation Battalion commander, a lieutenant colonel. The lieutenant colonel reported to the Army Aviation Brigade (TAAB) commander, a colonel. The colonel reported to the US Army Military District of Washington (MDW) / Joint Task Force, National Capital Region (JTF NCR) commander, a major general. The major general reported to the Army Chief of Staff, a four-star general. The PI reported directly to the 12th Aviation Battalion commander. Otherwise, her chain of command was the same.

B Company was a company-sized unit comprised of about 40 soldiers, including pilots, crew chiefs, and company leaders. The 12th Aviation Battalion was a helicopter battalion comprised of about 400 soldiers. The mission of the 12th Aviation Battalion was to support continuity of government operations in the NCR, as directed by US Northern Command. The battalion consisted of three active-duty helicopter companies (A, B, and C), a maintenance company (D), a headquarters company, and an attached (rotating) national guard helicopter company.

The 12th Aviation Battalion's parent organization, TAAB, included a US Army Priority Air Transport Battalion with fixed-wing aircraft at Joint Base Andrews-Bolling (JBAB), an Operational Support Airlift Activity Battalion with fixed-wing aircraft at DAA, an Airfield Division that operated the DAA airport and provided some air traffic control staff to the Pentagon Heliport, and a headquarters staff based at DAA.

7.2 Organizational Safety Program

TAAB's commander said that, per Army Regulation 600.20, commanders were responsible for everything their unit achieved or failed to achieve, and per Army Regulation AR 3850-10, commanders at every echelon were responsible for the Army safety program. He said that, prior to taking command, he had attended an eight-week pre-command course and he recalled that it included a one or two-day briefing by the US Army Combat Readiness Center (USACRC) that describe safety reporting tools of use to a brigade or battalion commander.

The commander relied on trained battalion safety officers to run the TAAB safety program. He had been working closely with a full-time GS-14 civilian safety manager posted at the brigade level since he assumed command. Their offices were in the same hallway at DAA, and they frequently interacted. A second brigade-level civilian safety manager had been hired shortly before the accident and began working two days before the accident. They had not yet had a chance to work closely

together, but the intent was for the additional safety officer to free up the existing safety officer for more aviation-related safety work. TAAB's commander said a challenge in managing TAAB's safety program involved the diversity of its activities. He said that prior to the accident, most of the safety issues the brigade safety program pertained to occupational health. When it came to aviation safety matters and reducing risk, he said that he looked primarily to the brigade standardization pilots, but if an accident occurred, he would pivot to his safety team.⁶⁵

TAAB's longer-tenured safety manager, a retired Army helicopter pilot, told investigators he advised, assisted, monitored, and managed various aspects of the brigade safety program. He said aviation was one of 14 program areas in the safety program required by Army Regulation AR 385-10, and he spent more time than he preferred on TAAB's ground safety and industrial hygiene program areas because the brigade did not have dedicated support in those areas. He said "well over 50 percent" of his time was devoted to aviation safety, but due to the mix of aircraft operated by TAAB, only some of that attention was devoted to helicopter safety.⁶⁶

TAAB's safety manager said TAAB convened semi-annual safety and occupational health advisory committee (SOHAC) and standardization meetings.⁶⁷ The TAAB commander stated that battalion commanders were the minimum audience at such meetings, but all available safety officers would attend or be dialed in to monitor the discussion. During SOHAC meetings, commanders and safety officers would discuss occupational health, ground safety, and flight safety-related issues that had been brought to their attention. The brigade commander would issue orders to the subordinate commanders and safety professionals about how to address issues that had been identified.

The 12th Aviation Battalion safety officer completed the US Army Safety Officer Course at Ft. Rucker (Ft. Novosel) in 2021. After that he served as a company safety officer at a unit in Kansas. He had been with the 12th Aviation Battalion and working in the safety officer role since August 2024.⁶⁸ He estimated that his position typically turned over about every three years. He reported spending about 75% of his work hours on occupational health and safety matters and 25% on aviation safety matters.⁶⁹

The battalion safety officer described his job at the battalion level as "managing the individual flight companies' safety officer programs."⁷⁰ In this role, he

⁶⁵ Interview with Col. Andrew DeForest, p. 61.

⁶⁶ Interview with Ross Steadman, p. 13.

⁶⁷ SOHAC meetings were required by Army Regulation AR 385-10 Army Safety and Occupational Health Program, July 24, 2023. Prior to July 2023, the Army referred to these meetings as a command safety council. Standardization meetings were required by AR 95-1.

⁶⁸ He was also an MBO.

⁶⁹ Interview with Benjamin Barczak, p. 45.

⁷⁰ Interview with CW3 Benjamin Barczak, p. 6.

coordinated quarterly battalion "safety stands" meetings, which involved discussion of safety issues brought to the attention of safety leadership by operational personnel and what to do about those issues. Examples they had dealt with included unmarked wires on their flight routes or high temperatures in the maintenance hangar.

He told investigators that he sometimes gave briefings at aircrew academics meetings organized by the standardization pilots. The last one addressed winter safety topics, such as avoiding slips and falls, using a buddy system in cold temperatures, an update on power line safety, and information about wildlife strikes. The battalion safety officer participated in SOHAC meetings twice a year, which he described as similar to the battalion safety stands meetings but, "just at a more centralized level with brigade leadership."

That battalion safety officer was responsible for both flight and ground safety, therefore it was his job to ensure that company safety officers submitted monthly packets of reports on required inspections of potential ground safety issues including fire, foreign object debris, chemical hazards, and noise.⁷¹ The battalion safety officer also facilitated an employee safety committee chaired by civilian employees from D Company (the maintenance company). At that meeting personnel could propose a change to a process they thought was inefficient or unsafe.

The battalion safety officer reported a good working relationship with TAAB's safety officer, stating, "as things are brought to my attention that can't be handled at my level, he's always more than welcome and willing to assist in pushing, elevating things higher to get some resolution behind them."

The battalion safety officer was asked if he had received safety reports about traffic conflicts, particularly in the vicinity of DCA. He said, "So since I've been here [about five months before the accident], I've had zero complaints or issues brought to my attention about the D.C. helo route corridor structures or any type of incidents along them with the exception being [a] laser incident." He added, "Every time I've flown on the DC helo routes, it's been a nonevent. It's been clear communication from tower and... good two-way communication between my aircraft and the other rotary-wing aircraft operating along the route structure so no issues to my knowledge."

The battalion safety officer was asked if any near-miss reports had surfaced following the accident. He said, "The only thing I would say that was brought to my attention post-accident was, I don't have all the details on it, but either the day before or a couple days before, one of our aircraft -- it was brought to my attention that commercial aircraft did a go around at DCA because of an aircraft operating on the

⁷¹ He stated that the Army was moving toward a centralized computer database for recording the results of these inspections in ASMIS 2.0.

corridor structure on Route 4.” He added, “There was no type of accusations that the helicopter was too high or doing anything, deviating from the standard operating procedures along the corridor structure. It was just one of those level of comfortability things I think is how it was explained to me.”

B Company’s safety officer had been with the company and working in that position since April 2024. He said that about 80% of his work hours were spent on occupational health and safety matters, and about 20% was spent on aviation safety matters.⁷² He reported that he had received one “near miss” type safety report since joining the company [about nine months before the accident]. It involved a close encounter with a bird that took place in July 2024.⁷³

7.2.1 Training of Safety Officers

The 12th Aviation Battalion safety officer, who had completed a five-week safety officer course at Ft. Novosel in 2021 described the course as “Kind of a crash course in all things safety, occupational health and the Army safety program.” He estimated that about half the course was focused on occupational health and safety and the other half was focused on aviation safety. B Company’s safety officer, who attended the course in early 2024, estimated that only about 1 week of the course was focused on aviation safety. A course outline provided to the NTSB on April 30, 2025, by the Army included a long list of topics. A condensed summary is provided below:

- ARAP
- FlightFax
- Unit safety program
- Safety meetings
- System safety
- Safety awards program
- Safety councils
- Risk management
- Psychology of mishap prevention
- SOP
- Occupational health
- Life safety code principles
- Range safety
- Motor vehicle mishap prevention
- Tactical safety
- Environmental safety

⁷² Interview with CW2 Zachary Fonda, p. 44.

⁷³ Interview with CW2 Zachary Fonda, p. 33.

- Ammunition and explosive safety
- Hazard analysis reporting
- OSHA standards (22 modules)
- Army safety and occupational health inspection
- Assessments, inspections and surveys
- Foreign object damage
- Aircraft landing site survey
- PE inspection
- Hazard management data analytics tool
- Fuel operations
- Operational hazard reports
- ASMIS 2.0 hazard management
- ARMS
- Aviation maintenance operations safety practices
- Army Mishap Investigations
- Mishap classification and reporting requirements
- Mishap reporting ASMIS PE
- Legal aspects of protecting safety requirements
- Pre-mishap plan
- Mishap site hazards and management
- Mishap investigation process
- Unmanned aircraft systems
- Evidence collection techniques
- Medical aspects
- Flight data recorder
- Metallurgy
- Mishap fires
- Evidence analysis techniques – Determining root causes
- Human Factors Analysis and Classification System
- Findings and recommendations
- Aviation safety trends
- Case studies

7.2.2 Safety Policy

TAAB's commander sought guidance for running the safety program from the following sources:

- Department of the Army Chief of Staff for Operations Plans and Training (G-3/5/7)
- The US Army Aviation Center of Excellence (USAACE)
- The US Army Combat Readiness Center (USACRC)
- Army Regulation AR 95-1, *Flight Regulations*
- A supplement to AR 95-1 that contained special regulations pertaining to the MDW
- Army Regulation AR 385-10, *The Army Safety and Occupational Health Program*

TAAB also maintained a brigade-level aviation and ground safety SOP.⁷⁴ The version in effect at the time of the accident was last revised in January 2024, following a major revision. TAAB's commander said he intended to publish a new version of the safety SOP that listed current staffing but decided to wait until an open MDW-level safety position was filled. That position was filled a few months later, but the accident further delayed the publication of a new revision.

TAAB's safety SOP defined the duties and responsibilities of commanders, staff, safety and occupational health/aviation safety officers, noncommissioned safety officers, supervisors, and individuals. It described procedures for risk assessment and risk management, prescribed monthly safety inspections, internal control audits, surveys, and hazard investigations, described the procedure for reporting mishaps, advised on safety education and training, provided for a safety awards program and defined the composition and responsibilities of the SOHAC. An excerpt from the TAAB safety SOP describing the safety related duties and responsibilities of TAAB personnel is reproduced below:⁷⁵

5. DUTIES AND RESPONSIBILITIES

a. Accident prevention and hazard reporting are the responsibility of all personnel. Policies, objectives, and standards must be established and clearly defined to ensure an effective accident prevention effort. Fulfillment of these responsibilities requires the complete cooperation of all members of the Brigade with specific job-related publications and procedures, all applicable regulations, and the unit SOP.

b. Brigade Commander. The Brigade Commander has overall responsibility for the safety and occupational health and accident prevention program within the

⁷⁴ Headquarter, The US Army Aviation Brigade. (January 17, 2024). The U.S. Army Aviation Brigade 12th Aviation Battalion, Airfield Division (Davison AAF), Operational Support Airlift Activity, and United States Army Priority Air Transport Aviation/Ground Safety Program Management Standard Operating Procedures,

⁷⁵ TAAB Safety SOP, Appendix 1, p. 7-1-2.

Brigade. The Commander will ensure all duties and requirements are established as necessary for the protection of personnel and equipment IAW AR 385 series.

c. Battalion Commander/Airfield Division Chief. The Battalion Commander / Airfield Division Chief has overall responsibility for the accident prevention program within their Battalion/Division. The Commander/Chief will ensure all duties and requirements are established as necessary for the protection of personnel and equipment IAW AR and DA Pam 385 series.

d. Company Commanders/Chief AFD/AFD section supervisors:

- 1) Assume overall responsibility for the accident prevention programs within their respective organization.
- 2) Implement accident prevention programs in support of this SOP.
- 3) Assign an officer on orders and appoint, on unit orders, a noncommissioned officer to assist in the management of the unit safety program N/A AFD.
- 4) Provide personnel with appropriate safety training for the positions held.
- 5) Appoint on unit orders the following program managers:
 - (a) Company Safety Officers N/A AFD
 - (b) FOD Prevention Officer/NCO if required.
 - (c) Hearing Conservation Officer/NCO
 - (d) Hazardous Material/ Hazardous Communication/ Hazardous Waste/Environmental Compliance Officer/ NCO
 - (e) Respiratory Protection Officer/NCO if required.
 - (f) Fire Prevention Officer/NCO (Fire Marshal)
 - (g) Master Driver
 - (h) Personal Protective Equipment Officer/NCO
 - (i) Safety NCO SEE AR 385-10 PARA 18-5, N/A AFD
 - (j) ALSE Officer/NCO if required
 - (k) Motorcycle Mentor if desired

e. Staff

- (1) Integrate risk management into all mission planning IAW ATP 5-19. Identify hazards most likely to result in loss of combat power, degradation of mission results, or mission failure.
- (2) Ensure a Safety Officer is included in the planning phase of all operations.

- (3) Integrate hazards and selected controls into functional area paragraphs, graphics and annexes of OPORDs and monitor implementation during execution.
- (4) Provide adequate resources to the safety staff.
- (5) Complete other duties as assigned in AR 385 series.

f. Safety and Occupational Health Specialist/Aviation Safety Officer (ASO). Responsible for the management of the commander's safety program to include the following:

- (1) Primary duty is to perform procedural analysis. SOH/ASOs will review the procedures personnel follow for equipment operations and the environment in which they must operate. These include determination of required tasks and sequence, exposure to hazards, criticality of each task and procedural steps involved, equipment characteristics, mental and physical demands, and the qualification of personnel available for these tasks.
- (2) Advises the Commander and maintains close liaison with Safety Representatives within the Brigade/Battalion/Company, Fort Belvoir, and Military District of Washington (MDW), on all matters pertaining to safety.
- (3) Serves as a member of the Brigade Safety and Occupational Health Advisory Council (SOHAC) and Standardization Committee.
- (4) Ensures that HAZLOGs are received quarterly from all BNs/AFD.⁷⁶
- (6) Participates in the quarterly pre-mishap plan review with the S3. Prepares a memorandum detailing the drill and maintains it on file in the safety office.
- (7) Performs other duties outlined in AR 385 series.

g. Safety Noncommissioned Officers AR 385-10 para 18-5. The aviation safety NCO assists, advises, and makes recommendations to the ASO on aviation mishap prevention matters. The aviation safety NCO will -

- (1) Maintain liaison with the command sergeant major, first sergeants, and other enlisted personnel on all aviation safety matters.
- (2) Observe aircraft support activities (such as petroleum, oil, and lubricants; maintenance; operations; and enlisted crewmembers'

⁷⁶ Item 5 in this list was missing from the original document.
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training) and participate in unit safety surveys and inspections to detect and report unsafe practices or procedures.
(3) Perform other USO duties prescribed in AR 385-10 paragraph 5-5.

h. Supervisors and Individuals.

- (1) Be responsible for personal safety, both on and off duty.
- (2) Be morally responsible to advise others about anyone who may, knowingly or unknowingly, be committing or about to commit an unsafe act.
- (3) Report all safety hazards to the chain of command, safety officer, or safety non-commissioned officer when on the spot corrections cannot be made.
- (4) Ensure all personnel receive continuous supervised training in job activities.
- (5) Report all personnel injuries, incidents and equipment damage immediately. Note: Report all first aid kit usage due to injuries in the workplace as a "near miss" event to the unit safety office.
- (6) Read and comply with all SOPs, policies and procedures.

7.2.3 Risk Assessment

TAAB's safety SOP stated:⁷⁷

Risk Management focuses on mission accomplishment while minimizing the potential effects of hazards that could cause loss of personnel, equipment, or impact mission success. Commanders, leaders, and managers at all levels will integrate risk management into all operations conducted by the TAAB."

a. Risk Management Procedure. The purpose of conducting risk management (RM) is to enhance the Army's primary decision-making process for identifying hazards and controlling risk across the full spectrum of Army missions, functions, operations, and activities. Risk Management will be conducted in accordance with DA Pam 385-30 and ATP 5-19.

TAAB's safety manager also stated that TAAB risk assessment requirements were spelled out in DA Pam 385-30. It defined a hierarchy of risk codes (RAC), with RAC1 being the highest. The safety officer said that RAC1 and RAC2 hazards were briefed at Military District of Washington SOHAC meetings. Lower-rated hazards,

⁷⁷ TAAB Safety SOP, p. 7-1-5.

such as those classified as RAC3 or RAC4, were handled at or below the brigade level.

The Aviation Risk Common Operational Picture (R-COP), developed by the US Army Aviation Center of Excellence, Aviation Center of Excellence (USAACE/AvCOE), was the principal risk management tool for TAAB flight operations. It could be customized to a certain extent by the user organization. The TAAB commander stated that USACRC revised the R-COP in 2023 to help commanders ensure the appropriate crew mix was selected for specific missions and to change the way that composite risk scores were calculated.

TAAB made its R-COP available to pilots as a computerized spreadsheet. Pilots used the R-COP for preflight risk assessment before all flight operations. The worksheet contained three main sections: crew, mission and environment, and weather. The user populated the spreadsheet with information about a proposed mission and the sheet calculated an initial risk score for each of these categories. Risk scores were classified as low risk (L), medium risk (M), high risk (H), or extremely high risk (Ext H).

The crew section incorporated current experience data derived from the latest CAFRS⁷⁸ download and assigned a risk rating based on crew member experience (time in the area of operations, accumulated flight time, 30-day flight time, flight time in the series of aircraft, NVD time, crew rest, etc.). The mission and environment section assessed risk for the type of mission (air movement, maintenance, external loads, etc.), power requirement considerations, training considerations (evaluation, readiness level progression, simulated instrument flight, etc.), terrain considerations (mountainous, over water, helicopter routes below 200 feet, etc.). The weather section evaluated such factors as winds, turbulence, thunderstorms, outside air temperatures, and ceilings. A fourth section asked the user to identify the greatest risk areas, allowing approvers to holistically identify the greatest risk to the flight and what could be done to mitigate it. Each section's original risk ratings were modified based on the proposed mitigation strategies to determine a final risk score.

The worksheet automatically populated risk scores for selected mission characteristics such as mission type, crew experience, and weather conditions, eliminating the need for the pilot to rate the risk themselves. A final Residual risk block was allocated to the Mission Briefing Officer (MBO) and Final Mission Authority Approval (FMAA) after they had done a detailed review of the mission and risk assessment with the IP or AMC. Residual risk could not be lowered by more than one level below the initial risk level.

⁷⁸ CAFRS – Centralized Aviation Flight Records System
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The pilot-in-command completed the R-COP and presented it to an MBO who reviewed it with the pilot. MBOs were specific officers designated by the battalion commander and were responsible for validating pre-mission planning according to appropriate regulations, standard operating procedures (SOPs), and ensuring that key mission elements were evaluated and briefed.

The pilot presented the reviewed R-COP to an officer who served as the final mission approval authority (FMAA). FMAAs were designated by the battalion commander. The R-COP risk assessment level determined the appropriate FMAA. For low-risk missions, company commanders could be the FMAA, moderate risk missions were approved by the battalion commander, high risk missions were approved by the brigade commander. Extremely high-risk missions were approved by the National Capital Region Task Force commander.

Brigade SOP stated that in-person MBO mission briefings were to be prioritized. If a verbal (telephonic) mission briefing was the only option, then an in-person FMAA briefing was required.

PAT 25's R-COP for the accident flight initially rated all three crew members as medium (M) risk. The IP was rated medium risk because he had less than 500 flight hours in the UH-60L and less than 100 NVD hours in the UH-60L. The PI was rated as medium (M) risk because she had less than 500 hours total flight time. The nonrated crewmember (NRCM/Crew chief) was rated as medium (M) risk because he had less than 500 hours in the UH-60L and less than 500 hours of NVD time in the UH-60L.

The following risk mitigations were identified to lower the crew risk rating to low (L) was, "IP is an IP with 400+ hours in LFA⁷⁹, NRCM, is FI with 350+ hours in LFA. IP and NRCM have flown together in the last 7 days." The Mission and Environment section was rated as medium (M) risk due to NVD terrain flight and helicopter routes below 200 feet. The following mitigation was used to reduce the risk rating to Low (L), "Crew will discuss CEFs operations and procedures. Crew will review route using sectionals and satellite data to verify hazards along the route of flight and establish CPs⁸⁰ at known hazards to help identify hazards, Night unaided flight will be along helicopter routes when cultural lighting becomes a hazard to NVG operations."

The Weather Complexity section was rated medium (M) risk due to winds greater than 30 knots, turbulence, and Moon illumination less than 25% or less than 30° above horizon or overcast. The Weather Complexity section used the following mitigations to reduce the risk rating to low (L), "Crew will have deliberate conversation involving lack of illum and turbulence. If turbulence is MOD crew will maintain 200 ft AGL during terrain flight at KNTG." The IP identified the greatest risk

⁷⁹ LFA - Local flight area.

⁸⁰ Check points

to the flight as “unknown hazards in the vicinity of 02VA and KHEF that could result in controlled flight into terrain,” The mitigation for this risk was “Crew will discuss hazards along the route and what techniques we will use to identify and avoid them. Crew will maintain 500 ft AGL along route of flight except KNYG range area and published DC Helo routes.”

The final overall R-COP risk rating for PAT 25 was low (L) for crew, mission/environment, weather, and residual. The PAT 25 R-COP was signed by the battalion standardization officer as the MBO and the Bravo Company commander as the FMAA. Copies of PAT 25’s R-COP and mission brief and approval form are contained in Attachment 4.

7.2.4 Safety Assurance

TAAB used various tools and processes for safety assurance. These included in person meetings, safety reports, audits, surveys, and tracking systems. The TAAB commander said if air crew members encountered a hazard in flight, they were likely to discuss it with their MBO afterward. That information would filter up through the safety chain of command and be discussed at a SOHAC meeting. In addition, TAAB’s commander touched base weekly with battalion commanders, at which time safety issues might be discussed. Pilots could also submit occupational hazard reports (OHRs) to their safety officers.⁸¹

According to the TAAB safety manager an Army Safety Management and Information System (ASMIS) hazard reporting module had been introduced by USACRC in 2021. Any TAAB member could enter an occupational hazard report (OHR) using the system for safety officers to address, and they could even submit one anonymously if they wished. However, the safety officer said he had not received an OHR “in a while.” He said that one of the fixed-wing units was good about entering hazard reports, but the other battalions had been slow to adopt the practice. As a result, he had begun having units present their near-miss reports at SOHAC meetings to encourage use of the system. He said that near-miss reports had been discussed at such meetings in June and November 2024.

TAAB personnel had the option of reporting near-misses using the Army Safety Management and Information System (ASMIS), a system developed and operated by the USACRC. Reports of class A or B mishaps were copied to TAAB’s commander. TAAB’s commander said that prior to the accident, he did not routinely discuss such reports with the USACRC, but he had done so since the accident.

In some cases, the FAA would notify the Army about a possible pilot deviation through a Department of the Army Representative and such reports would be

⁸¹ OHRs were forwarded to the U.S. Army Aeronautical Services Agency at the Headquarters, Department of the Army (HQDA).

forwarded to the appropriate battalion commander to investigate. The battalion commander could choose to elevate the report for a more formal investigation at the brigade level.

The TAAB commander stated that the Army administered an annual survey to brigade personnel through the Army Readiness Assessment Program (ARAP). A US Army web site stated that the ARAP survey was a “web-based survey to assess a unit's safety culture and readiness” that helped commanders “identify and address potential issues related to safety and risk management.”⁸² The TAAB commander said the ARAP survey included open-ended items, and those comments were forwarded to battalion commanders and safety officers. The TAAB commander said he found the results of helpful for monitoring the brigade’s safety culture.

The TAAB commander said that the most recent ARAP survey had been completed in the fall of 2024. Concerns reported by 12th battalion personnel involved their use of older helicopters with frequent maintenance issues to perform a “no-fail mission every day” while also providing enough flying hours to conduct adequate training. This was very challenging. The TAAB commander said that a retiring brigade instructor and standardization pilot had communicated concern to him about the declining experience levels of brigade pilots, which made it more difficult to crew missions with the right mix of personnel.

Every three years, TAAB participated in an Aviation Resource Management Survey (ARMS) inspection conducted by US Army Forces Command (FORSCOM). This inspection included items related to TAAB’s safety program. The most recent ARMS inspection of TAAB’s 12th Aviation Battalion was conducted between November 18-20, 2024. The inspection evaluated the 12th Aviation Battalion aviation safety management as 94% compliant, or “green.”

TAAB’s safety officer could not recall any actionable safety items identified through the most recent ARMS or ARAP surveys, aside from some recordkeeping issues. However, TAAB tracked hazards identified by an ARAP, ARMS, and other sources using a module of the ASMIS.⁸³ TAAB’s safety officer said that a hazard tracking module had been developed about three years before the accident, but TAAB had only begun using it about 6 months ago so that commanders could more easily track hazards. According to the TAAB safety officer, the delayed use of this system was due to a slow culture change and limited staffing.

TAAB’s safety officer said an expired guidance document titled DA Pam 385-90, *Aviation Safety Program Management*, had been a source of safety guidance for aviation units prior to July 2023, at which time the new AR 385-10 and DA Pam 385-

⁸² Information from earap.safety.army.mil, downloaded May 26, 2025.

⁸³ TAAB’s safety officer stated that prior to that, hazards were tracked using spreadsheets.

10 were published, and DA Pam 385-90 was phased out. Some processes prescribed in DA Pam 385-90 were not carried forward in the new guidance materials. A noteworthy omission was the removal of a requirement to perform an annual self-assessment using ARMS checklists that was referred to as an aviation accident prevention survey (AAPS). When the AAPS requirement was phased out, it was removed from the TAAB safety SOP.

TAAB's safety officer said AAPS surveys could identify problems like operations not maintaining flight records or maintainers missing critical tasks. He said many US Army safety professionals wanted to reinstate AAPS, and that he and TAAB's commander had decided to reinstate AAPS as a TAAB requirement in the next version of the brigade safety SOP.

TAAB's safety officer said TAAB had begun implementing the DoD Safety Management Center of Excellence's Army Safety and Occupational Health Management System (ASOHMS) "a few years ago." He described ASOHMS as "a safety management system". The Army had published a directive in 2024 requiring the implementation of ASOHMS. ASOHMS implementation had three stages. TAAB's safety officer said TAAB had progressed from stage 1 (having safety policies and procedures in place) to stage 2 (verifying compliance with existing policies) in May 2023. They were working to undergo a stage 2 audit in the near future to reach stage 3 (verifying the effectiveness of existing policies). The safety officer expected that stage 3 would entail greater focus on safety assurance processes. TAAB's safety officer said that USACRC provided a web portal where he could monitor various analytics, such as Army aviation mishap statistics.

TAAB's commander was asked what mechanisms the 12th battalion had to identify exceedance of certain flight parameters, such as flying above the maximum altitude on a DC helicopter route. He said the Army might be informed about such an event if the FAA notified them about an alleged pilot deviation, however, the Army did not have a technical mechanism to remotely identify such events. TAAB's safety officer, asked the same question, said TAAB would only learn about such an occurrence if a crew was "violated by the FAA and we had to answer that or if the crew self-reported." TAAB personnel said they did not have a program to monitor helicopter flight operational data and analyze it for safety assurance purposes.

TAAB's safety officer was asked what aspects of TAAB's safety program could identify near midair collisions between military and civilian aircraft in the DCA terminal area, assess the risk of such events and develop risk mitigation strategies for de-conflicting helicopters and civilian airliners operating out of DCA. He said, "If people had been filing near-miss reports, it would -- would've helped us gather that there were areas of concern of loss of separation."⁸⁴

⁸⁴ Interview with TAAB Safety Manager, Ross Steadman, p. 41.
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7.2.5 Safety Promotion

TAAB's safety manager said that TAAB safety promotion activities included safety impact awards presented during safety stand downs. TAAB's safety manager was asked what steps it had taken to promote safety reporting, and he said they did the right thing to make sure people were not afraid to report and they were not punished for reporting safety concerns. TAAB's commander said the USACRC published a monthly newsletter called FlightFax that was distributed to TAAB personnel. It contained articles on safety topics specific to Army aviation.

7.2.6 Post-Accident Safety-Related Changes

The TAAB commander outlined post-accident changes that the Army either had made because of the accident or intended to make because of the accident. These included the following:

- Verifying that ADS-B was working before every flight
- Elevating the risk score and the signoff level for any flight where ADS-B will be turned off.
- Requiring Blackhawk crew chiefs to open their windows during takeoff and landing to improve visual monitoring.
- Modernization of the 12th aviation battalion helicopter fleet by the fourth quarter of 2026.

7.3 Pilots' Perceptions of Organizational Culture

A sampling of personnel at various levels of the TAAB were asked about their perceptions of safety culture. The wife of the CE on the accident flight said B Company's mission was always prioritized, even if enlisted personnel like her husband had to work excessive hours on an unpredictable schedule and started to show signs of burnout like feeling exhausted, visiting mental health facilities, or downing themselves to stop flying. She said if CEs were burned out, they might not perform a maintenance task to standard or record the work correctly. She was concerned appropriate action might not have been taken at a higher level if things had started to slip in that area. She wondered if the battalion leadership was aware that CEs like her husband were displaying signs of burnout.

A pilot in B Company said, "I've never been in a unit like this before, so -- a lot of long days sitting at work. But that is the culture, generally speaking. I think everyone gets along fine. There's no -- like we don't have any bad seeds or like -- so like the group is tight knit, but what we do just -- it's just long days." Asked specifically about safety aspects, he said he thought people in the unit knew what they did was dangerous and they took it seriously. He said that if he ever had a safety concern, he would have no reservations about approaching battalion leadership. Another pilot in

B Company thought the safety culture in his unit was "pretty high." He added, "There's not... any appetite for, like, doing things unsafe. If you're found to do something unsafe, you are immediately corrected." He found B Company to be "very professional" and the pilots did not "work beyond what [they] were able to do."

A B Company standardization pilot said he felt the 12th Aviation Battalion culture was very professional. He thought most crew were experienced, and he believed that this contributed to a level of professionalism. A C Company standardization pilot said it was his perception that safety was the number one priority in the 12th Aviation Battalion. He said they had pulled back on risk acceptance at the direction of the battalion commander because they had junior crews, and he said safety culture in the Battalion was "really, really big." He also felt the battalion was highly professional. He said their budget was "pretty low" so they had to make up for that with "training and personnel".

A pilot in the 12th Aviation Battalion who served as a flight operations officer-in-charge was asked about the culture of the unit, compared to others in which he had served. He said, "It's stateside and everyone is, you know, with their family, or living domestically here versus in a deployed environment or -- so I think people kind of have their own lives a little differently outside of work. However, it seems like a pretty good unit overall. I really enjoy working with everyone there and I enjoy the types of flying that we do."⁸⁵

The 12th Aviation Battalion Commander said the safety culture in the battalion was good and she had made safety a priority. She encouraged crew members to speak up if another crewmember was underperforming. She stated, "62 percent of the formation is pointed against a mission line every single day, and so while we don't have those field exercises, while we don't have those overseas deployments, that becomes tough in its own way." She added, "That's 12 hours every day that you are tied to those lines, and I think that can be a strain on people. That can be a strain on families before you start to... feel that as an impact." She said safety could be accomplished through her philosophy, "discipline plus standards equals safety." To her, that meant doing the right things, the right way, every time.

A battalion level standardization pilot said the 12th Aviation Battalion had a "very good safety-oriented culture." He added, "Our battalion commander basically, what we call her bumper sticker, and she puts it on every slide, every academic we do, is 'discipline plus standards equals safety'. You'd be hard pressed to find somebody that's been in the organization that can't recite that by memory. The mission we have itself is not inherently dangerous. It's just air movements of personnel. That is our unit's mission. So we don't do a whole lot of tactical mission stuff here where the risk will really start to creep in. I feel that we operate pretty safely

⁸⁵ Interview with CPT Noah Sawatzky, p. 27.

as an organization.” He thought morale had been “normal” with workload “a little bit higher” than usual because of the swapping out of the attached National Guard company for a new one from a different state.

The battalion-level standardization pilot thought most soldiers were happy to be in TAAB because they were not deploying. He said that about a year before the accident, the unit received approval from US Army Human Resources Command to not have to accept pilots right out of flight school, and as a result they had been attracting more experienced aircrew. However, some of the people they were attracting were expecting an easier time and they were surprised that they were required to be on duty 12 hours a day, two or three days a week.⁸⁶

8.0 Additional Information

8.1 US Government Accountability Office (GAO) Oversight

A 2023 GAO report noted deficiencies in Army aviation safety management, including the absence of a system for tracking the status of accident investigation recommendations and a lack of continuous and systematic evaluation and updating of risk management worksheets. The report noted deficiencies in Army National Guard safety management including high workload that degraded Army National Guard safety officer efforts and National Guard pilots’ challenges meeting flight hour goals.⁸⁷ Army aviation conducted a safety “stand down” in late 2023, during which units paused missions and training flights to discuss safety issues. By March 31, 2024, the Army experienced twice the usual number of serious mishaps and fatalities in the first half of the fiscal year. In April 2024, the Army ordered a “safety stand up” during which units received additional training on safety-related procedures and protocols.⁸⁸ By September 30, 2024, Army Aviation had experienced 17 class A mishaps (accidents that are fatal or cause permanent disability or more than \$2.5M in damage) during the full 2024 fiscal year, the most since 2007.⁸⁹

8.2 DCA-Area Helicopter Routes

Helicopter routes for the Washington, DC area were graphically depicted in an FAA publication titled Helicopter Route Chart (Baltimore-Washington).⁹⁰ Helicopter

⁸⁶ Interview with CW4 Matthew Ingalls, p.27.

⁸⁷ US GAO. (2023). National Guard helicopters: Additional actions needed to prevent accidents and improve safety. Report No. GA-23-105219, March 2023.

⁸⁸ South, T. (2024). Army pushes more safety training as helicopter crashes spike. *The Army Times*, April 10, 2024.

⁸⁹ Myers, M. (2025). Army aviation was having a bad few years even before Wednesday’s crash. January 31, 2025.

⁹⁰ Effective date 26 December 2024 – 20 February 2025.

routes were designed to enhance helicopter access and operation within high density traffic areas. The *Description of Routes and Altitudes* section of the chart described paths helicopters should follow with reference to key landmarks, with maximum altitudes expressed in terms of height above mean sea level (msl).

A battalion standardization pilot from the TAAB stated that although transition zones were available, helicopter routes were the battalion's preferred way to train their pilots in and around Washington, DC, and that they stuck to the helicopter routes as much as possible.⁹¹ However, he stated that if he was providing an NVG APART to another pilot, he would not do it in that area. He would do it between DAA and Quantico.

According to the ATC group, which received a briefing from an FAA charting subject matter expert, the DC helicopter routes do not have any lateral or vertical dimensions besides what was depicted on the chart. Pilots were expected to review the visual depiction of the routes and follow the lines. Aircraft could transition between an operating zone and a helicopter route at any point along the route with an ATC clearance. There were compulsory and non-compulsory reporting points along the route where aircraft position reports were provided to ATC. These points commonly served as holding points.

Proceeding southbound along the northern portion of Helicopter Route 4, 12th Aviation Battalion pilots said that they would fly over the Potomac River while hugging the east bank. They maintained a distance from the shore that was sufficient to avoid land terrain and obstacles, unless a traffic conflict required further lateral deviation (if vertical clearance would not be adequate with the low altitude buffer). There was no common agreement among pilots as to exactly how much they could deviate laterally from the perceived centerline of the route, except that they would deviate enough to visually deconflict with other traffic.

The Cabin John non-compulsory reporting point, which corresponded to the American Legion Bridge, was the northwesternmost end of Helicopter Route 1. Proceeding eastbound, the beginning segment between Cabin John and Chain Bridge had a published maximum flight altitude of 1300 ft msl. The maximum altitude of the next segment between Chain Bridge and Key Bridge was 700 ft msl. Maximum altitude between Key Bridge and the Memorial Bridge was 300 ft msl. From Memorial Bridge to Hains Point, the max altitude was 200 ft msl. The western portion of Route 1

⁹¹ Interview with CW4 Matthew Ingalls, p. 10.

terminated at Hains Point, where the eastern half of the route continued up the Anacostia River.

Hains Point was the northernmost point of Helicopter Route 4, and the segment between Hains Point and the Wilson Bridge also had a maximum altitude of 200 ft msl. The maximum altitude was 300 ft msl over the Wilson Bridge. The segment between Wilson Bridge and Broad Creek was 600 ft msl. The segment from Broad Creek ending at Fort Washington had a max altitude of 1000 ft msl.

TAAB had revised its SOP to restrict the minimum flight altitude to be no lower than 100 ft msl from the maximum route altitude for noise abatement. Most pilots interviewed by the helicopter operations and human performance group stated that they would typically fly below the 200 ft msl while on Helicopter Route 4 with some buffer (e.g. 50-100 ft) by referencing the barometric altimeter to strictly avoid exceeding the maximum altitude, however some said they would fly at 200 ft msl.

Traffic landing on DCA runway 33 crossed Route 4 approximately 3/4 of a mile south of the Bolling compulsory reporting point. Pilots interviewed reported that seeing traffic landing on runway 33 was not typical. Pilots who had encountered fixed-wing traffic landing on runway 33 reported that they had been instructed by ATC to hold until the traffic had landed, regardless of whether they had the traffic in sight. Pilots stated that Hains Point and what was commonly referred to as “the golf balls” (Figure 47) were used as holding points north and south of the runway 33 approach area, respectively. Interviewed pilots were under the impression that there would be enough vertical clearance with the landing traffic on Runway 33 if the aircraft stayed at or below the max published altitude of Route 4.

DESCRIPTION OF ROUTES AND ALTITUDES

ALL ROUTES MAY BE ALTERED AT PILOT'S REQUEST OR AS DIRECTED BY ATC.

ROUTE ALTITUDES ARE MAXIMUM

- RT. 1: AMERICAN LEGION BRIDGE OVER POTOMAC RIVER, EAST OF ROOSEVELT ISLAND TO THE TIDAL BASIN. OVER WASHINGTON CHANNEL TO ANACOSTIA RIVER. NORTHEAST OVER ANACOSTIA RIVER TO RIVERDALE AND VIA BALTIMORE WASHINGTON PARKWAY TO GREENBELT.
- ALTITUDES: AMERICAN LEGION BRIDGE AT OR BELOW 1300 FEET MSL, CHAIN BRIDGE AT OR BELOW 700 FEET MSL, KEY BRIDGE AT OR BELOW 300 FEET MSL, MEMORIAL BRIDGE AT OR BELOW 200 FEET MSL NOT ABOVE 200 FEET MSL UNTIL JAMES CREEK MARINA, AT OR BELOW 300 FEET MSL TO 11TH STREET BRIDGE, AT OR BELOW 500 FEET MSL TO PENNSYLVANIA AVENUE, AT OR BELOW 700 FEET MSL FROM PENNSYLVANIA AVENUE TO RIVERDALE, AT OR BELOW 1300 FEET MSL TO GREENBELT. (HELICOPTERS CROSSING POTOMAC RIVER TO OR FROM THE PENTAGON SHALL BE AT OR BELOW 200 FEET MSL).
- RT. 4: FORT WASHINGTON OVER POTOMAC RIVER TO WILSON BRIDGE. THEN VIA EAST BANK OF POTOMAC RIVER TO ANACOSTIA RIVER. INTERCEPT ROUTE 1 AT ANACOSTIA RIVER.
- ALTITUDES: AT OR BELOW 1000 FEET MSL AT FORT WASHINGTON, DESCEND TO 600 FEET MSL ABEAM BROAD CREEK INLET, BEGIN DESCENT FROM 600 FEET MSL TO ARRIVE AT 300 FEET MSL OVER WILSON BRIDGE, THEN AT OR BELOW 200 FEET MSL NORTH OF WILSON BRIDGE.

Figure 44. Description of helicopter routes and altitudes from the helicopter route chart.



Figure 45. Approximate location of the accident crash depicted in red circle over the Helicopter Route
Chart: Baltimore-Washington

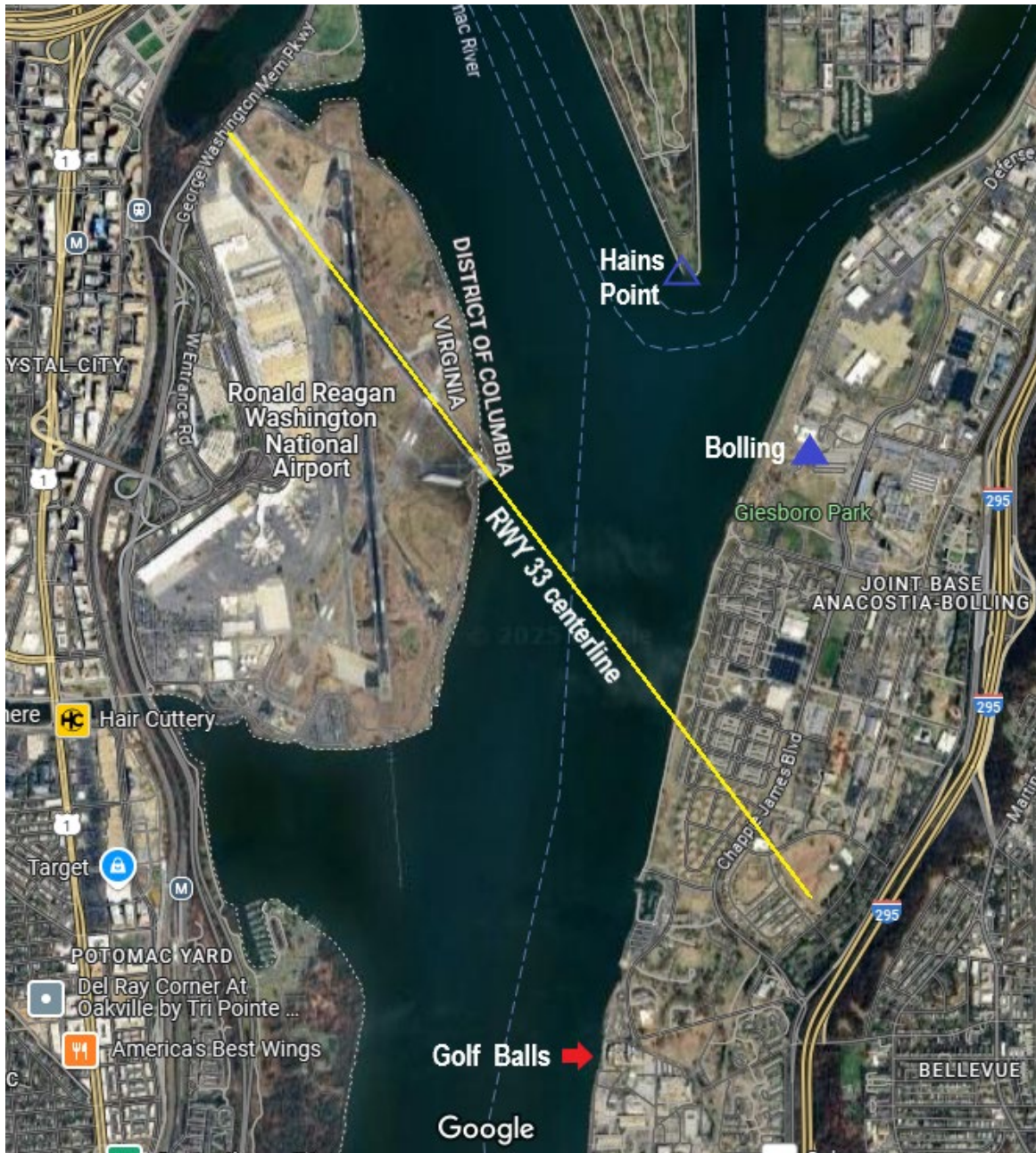


Figure 46. Satellite view of Google Maps with geographical references

8.3 Use of Barometric Altimeter and Radar Altimeter

All altitudes specified in FAA aeronautical charts were presumed to be in feet msl unless noted otherwise. The barometric altimeter was the primary flight altitude reference instrument for aircrews. When operating below 18,000 feet, aircrews were required to set the current local altimeter setting.⁹² When setting the altimeter during

⁹² 14 CFR Part 91.121
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the before flight checks, with the current altimeter setting dialed in, the altimeter was to read ± 70 feet of the published field elevation.⁹³

The radar altimeter (radalt) used radio waves to measure the distance between the aircraft and the terrain or water below the aircraft. The radalt provided feet of altitude above the ground. Army aviators use the radalt as their primary source of altitude information when performing low level terrain flight (below 200 feet) and for obstacle avoidance.⁹⁴

There were two adjustable settings (bugs) on the radalt where pilots could set lower and upper altitude limits. Typically, these are set at 80 feet for the lower limit and 800 feet for the upper limit.⁹⁵ That was standard practice taught in Army flight school. However, pilots could adjust those bugs as required, depending on the operational environment.

8.4 Interviews with Other DC Area Helicopter Operators.

There were numerous organizations that operated helicopters in the Washington DC area regularly. The major operators were the United States Coast Guard (USCG), United States Air Force (USAF) 1st Helicopter Squadron, US Army 12th Aviation Battalion, United States Marine Corps (USMC) HMX-1, Maryland State Police, US Park Police, Washington DC Metropolitan Police, Fairfax County law enforcement and medivac, Prince Georges County Police, Stat-Med, and MedStar. The USCG, 1st Helicopter Squadron, and Prince Georges County Police agreed to be interviewed regarding their operations.

8.4.1 USCG NCRADF Detachment⁹⁶

The USCG National Capital Region Air Defense Facility (NCRADF) helicopter detachment operated about 4 MH-65E, Dolphin helicopters out of their facility at DCA. Their mission was to intercept, identify and escort, low slow targets of interest identified by Continental NORAD in support of the Noble Eagle mission. All their pilots were experienced, with greater than 1,000 flight hours. USCG helicopters used the DC area helicopter routes regularly with at least two sorties per day. Generally, when DCA was landing on both northerly runways (runways 01 and 33) they tried to stay clear of the helicopter routes that went past DCA. When DCA was using runway 33 it was very common for the DCA tower to either deny the request to use Helo Route 4 or issue VFR holding instructions. Helicopter crews frequently initiated a

⁹³ TM 1-1520-237 MTF (maintenance test flight). The FAA AIM 7-2-3;a(3) states that the altimeter should be ± 75 feet of the field elevation.

⁹⁴ CW3 Halterman interview transcript

⁹⁵ CPT Sawatzky interview transcript

⁹⁶ USCG interview transcript.

request to use visual separation when traffic was pointed out by the DCA tower. USCG pilots interviewed expressed the opinion that visual separation was a useful tool that allowed efficient use of the airspace.

Internally, the USCG detachment had an SOP that stipulated that when on a charted route they should fly the maximum published altitude, and when flying below 300 feet in the NCR, airspeed should be less than 100 knots, and the landing lights should be ON. Additionally, it was policy to monitor commercial traffic and avoid flying between the bridges (Memorial Bridge and Wilson Bridge) if there was a high volume of traffic. For active air defense and law enforcement missions it was policy to operate with ADS-B Out turned OFF. Aircraft that were on training flights and not on alert status, were to operate with ADS-B Out turned ON.

Close calls or near miss reporting was done internally within the USCG safety program. Most traffic conflicts the USCG pilots experienced involved other helicopters. Pilots could use one of the FAA's voluntary reporting systems such as the ASRS or ASTP, but it was not part of the safety program to submit reports into those systems. The detachment did not have a flight operations quality assurance (FOQA) program that reviewed operational flight data for trends or exceedances.

The choice to wear NVGs at night was up to individual crew members. There was no requirement for all crew members to wear NVGs at the same time. Green phosphor NVGs were the type used when operating in the NCR. The helicopters were equipped with TCAS which provided information on the multifunction displays in the cockpit. The TCAS did not provide traffic advisories (TAs) or resolution advisories (RAs), but it was used to enhance aircrew situational awareness. Crews also used iPads with Foreflight. The iPads were equipped to receive cellular data, and Stratus devices were used to receive ADS-B data and transmit it to the iPad, which allowed current flight and traffic data to be displayed on the iPad.

8.4.2 USAF 1st Helicopter Squadron⁹⁷

The USAF 1st Helicopter Squadron operated 22 UH-1N Huey helicopters based out of Joint Base Andrews. The squadron's mission was distinguished visitor lift in the NCR and some contingency response for Air Force District of Washington and Joint Task Force NCR. Half their pilots were experienced aviators with over 1,000 hours each, the other half of the pilots were on their first tour having recently graduated from flight school with about 150 flight hours each. The squadron used the Baltimore-Washington helicopter routes on a daily basis, with no less than 8 flights a day specifically using Helo Routes 1 and 4. They flew the maximum altitude allowable on any particular route, and per USAF guidance the flew no lower than 100 feet agl.

⁹⁷ 1st Helicopter Squadron interview transcript
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There were times when DCA tower would request that the squadron limit its use of those routes between the times of 2100 to 2300. USAF pilots reported that encountering landing traffic to DCA runway 33 was occasional when on Route 1 and 4. There were instances when they entered visual flight rules (VFR) holding unprompted and inform DCA tower. Generally, they followed ATC guidance and initiated a visual separation request if they saw traffic ahead of them. For any flight after sunset, it was standard procedure to fly with NVGs worn by all crew.

For all operations, both mission related and training, it was policy to operate with ADS-B Out turned OFF. The crews used iPads with Foreflight installed, which received ADS-B In data. The squadron did not have a FOQA program that reviewed operational flight data for trends or exceedances. Their helicopters were not equipped with TCAS. If the flight crew determined they might have had a near miss or close call, that was reported using a hazardous air traffic report (HATR) and inputted into the Air Force Safety Automated System (AFSAS). The AFSAS could only be accessed by the Wing Chief of Safety and the Air Force Safety Center. Flight crews could also submit a SAFEREP which was a voluntary Air Force reporting system through which airmen could report aviation and ground safety hazards. Safety information was disseminated during quarterly safety meetings. There was an annual review where mishaps were grouped, labeled, and a report was produced for unit commanders and installation commanders. There had been no HATR or other aviation safety report for any flights using Route 1 and 4 over the last few years.

8.4.3 Prince Georges County Police⁹⁸

The Prince Georges County Police operated four MD-520N NOTAR helicopters out of College Park Airport, Maryland. The primary mission was to provide aerial support for various law enforcement missions including assisting ground units during pursuits, aerial surveillance, and search and rescue operations. Eighty percent of their operations were flown at night. The aviation branch consisted of three pilots who had a combined average of over 5,000 hours of rotorcraft flight time. Two of the pilots had been with the unit for over 20 years and the third for 10 years. The flight crew consisted of a pilot and a tactical flight officer (TFO) who was not a rated pilot. They use the Baltimore-Washington helicopter routes about once every three days. It provided a quick way to get to National Harbor from College Park. Regarding operations near DCA they knew "rush hour" for DCA was between 1600 and 2100 and that on Sundays it was really busy, so they generally stayed off routes 1 and 4, and used zone transitions instead. Their SOP prohibited flying under any other aircraft, and when on route 4 and when there was landing traffic at DCA within 2 miles of them, they would turn east or reverse course until clear of the traffic. Additionally,

⁹⁸ Interview with Lt Aaron Smith, Chief of the Aviation Division, Prince Georges County Police.
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when on route 1 and 4 when DCA had southerly landing and departing aircraft, if an aircraft was within 2 miles they would hold to avoid wake turbulence.

Close calls or near misses were tracked informally within the law enforcement community. Most of these events involved other helicopters flying through their orbit when they were on active law enforcement operations. These instances would be shared with other operators via email. They operated with ADS-B Out turned ON except when on certain law enforcement operations where broadcasting their position affected their ability to interdict criminal activity. However, it was SOP for them to have ADS-B Out turned ON whenever there was other traffic within three miles of their position for situational awareness and a safety precaution. They mainly monitored ATC frequency to determine where other traffic is and how close. ADS-B In data was displayed on the helicopter's multifunction display (MFD). Typically, they did not fly with iPads unless they were on a cross-country flight.

Most of their flights were at night and they used ANVIS 9 NVGs when flying at night. The ANVIS 9s tended not to have the halo effects around point lights and tended not to shut down when over-saturated with bright lights. Their aircraft were also equipped with helicopter terrain warning system (HTAWS) and synthetic vision which they found extremely useful for avoiding objects. The unit did not have a FOQA program that reviewed operational flight data for trends or exceedances.

Report submitted by:

William J. Bramble Jr., Ph.D.
National Resource Specialist (Human Performance)

Van S. McKenny, IV
Aerospace Engineer (Helicopters)